Windshear Detection

How MODE-S Wind finds low-level windshear from the aircraft's own data, and how to read the Windshear page

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

In short. Every aircraft on final approach to Helsinki-Vantaa answers the Mode S interrogations of the ground stations with its true airspeed (TAS) and ground speed (GS). Their difference, TAS − GS, is the headwind the aircraft is flying into, measured about every two seconds. MODE-S Wind follows this headwind along each aircraft's path and looks for two things:

- a **headwind ramp** — a sudden loss or gain of headwind along the path of one aircraft, scaled to a 1-km equivalent so that a quick change counts more than a slow one (Section 3.4);
- a **change between flights** — the wind of the aircraft in each 200-ft height band compared with all landings and departures of the previous 45 minutes, which shows a front or a gust front arriving (Section 3.6).

The result is graded like the alerts of airport windshear systems: ADV advisory from 10 kt, WSA windshear alert from 15 kt (with a real headwind change of at least 14 kt) and MBA microburst alert from a 30 kt loss. The detection runs on the server for every landing and departure, whether or not anyone has the page open; the Windshear page shows the alerts, a 24-hour alert log and a windshear report for every landing (Section 5).

An experimental research product. The alerts come from a passive receiver and describe air that an aircraft has *already* flown through. They are not an operational windshear warning, they are not certified, and they must not be used for flight operations or air traffic control. Use them to study the low-level wind at the airport and to compare with official reports.

Contents

1 What windshear is **2**

2 What the aircraft tell us **2**

3 The detector step by step **3**

3.1 Cleaning the replies . . . 3
3.2 Distance along the path . . . 3
3.3 Smoothing . . . 4
3.4 Headwind ramps . . . 4
3.5 Horizontal F-factor . . . 5
3.6 Change against the earlier flights (band comparison) . . . 5
3.7 Levels . . . 6
3.8 From an event to an alert . . . 7
3.9 A windshear report for every flight . . . 7

4 The scientific basis **7**

5 The Windshear page **8**

5.1 Banner, flight strips and ILS profile . . . 9
5.2 The alert log . . . 10
5.3 The alert message . . . 11
5.4 The engine status line . . . 12
5.5 Approach History: the WS view . . . 13
5.6 The other panels . . . 14

5.7 Turbulence (EDR) 19
5.8 The boundary layer: low-level jet and surface inversion 20
5.9 METAR and pilot reports 21

6 Cases from October 2026 **22**

7 Limitations and good practice **23**

8 Quick reference **25**

1 What windshear is

Definition. Windshear is a change of the wind speed or direction over a short distance. For aviation the dangerous part is *low-level* windshear: ICAO defines the wind shear warnings of an aerodrome for the layer between the runway and 500 m (1 600 ft) above it, along the approach and take-off paths [1]. What matters to an aircraft is the change of the wind *along its own path*, in particular of the headwind: a headwind change of 15 kt or more, lasting longer than a gust, is usually regarded as significant [2, 3].

Why a loss of headwind is the hazard. An aircraft flies through the air, not over the ground. If the headwind suddenly drops by 20 kt, the airspeed drops by the same amount at that instant, while the ground speed stays the same. On a stabilised approach at 140 kt the wing then has 20 kt less airspeed and less lift, the aircraft sinks below the glide path, and the crew must add thrust and pitch up to recover. A **gain** of headwind does the opposite — the aircraft balloons above the path and speeds up; it is less dangerous but often comes just before a loss (the typical sequence in a microburst: headwind gain, downdraft, tailwind). Loss is therefore treated as the more serious direction, and only a loss can become a microburst alert.

The F-factor. NASA and the FAA combined both effects — the change of the horizontal wind and the downdraft — into one hazard index, the *F-factor* [4, 5]:

$$F = \frac{\dot{W}_x}{g} - \frac{w}{V}, \tag{1}$$

where $\dot{W}_x$ is the rate of change of the headwind *loss* along the path (m/s^2; a tailwind increase counts as positive), g the gravity, w the vertical wind (negative in a downdraft) and V the airspeed. F is the loss of the aircraft's climb capability expressed as a gradient: $F = 0.1$ means the shear takes away the equivalent of a 10 % climb gradient. Averaged over about **1 km** of path it best separates hazardous from harmless shear; the FAA considers windshear hazardous at $F > 0.1$ and uses 0.13 as the "must alert" level of airborne windshear systems [5].

What happens at Helsinki-Vantaa. Downbursts from thunderstorms occur in Finland mainly in summer. More common causes of low-level windshear at EFHK are *fronts* and *gust fronts* that change the wind within minutes, *low-level jets* and *inversions* that make the wind turn and increase sharply over a few hundred feet (often at night and in the morning), and strong, gusty winds in autumn and winter storms. The detector is built for all of these, not only for microbursts.

2 What the aircraft tell us

Mode S Enhanced Surveillance. Ground stations interrogate each aircraft for registers that its transponder fills from the flight computers. In Finland they are mainly the transmitters of Fintraffic's nationwide *wide-area multilateration* (WAM) network [11, 12], together with Mode S radars; WAM is not tied to the rotation of a radar antenna and can ask an aircraft as often as it is configured to. MODE-S Wind receives the replies passively and cannot change how often they come; the Statistics page shows how often each register is answered at the moment, on final, in the terminal area and en route. Two registers matter here:

- **BDS 5,0** (track and turn report): roll angle, true track, ground speed, track rate and **true airspeed** — all from the same instant;
- **BDS 6,0** (heading and speed report): magnetic heading, indicated airspeed, Mach and the vertical rates.

On final approach at Helsinki-Vantaa an aircraft is asked for its BDS 5,0 register about every two seconds, and a new value of TAS − GS arrives about every four seconds, or every 290 m of path [17] (about a quarter of the replies repeat the previous content, because the register has not changed or not yet been refreshed).

ADS-B ground speed. In addition, every aircraft broadcasts its ground speed and track about twice a second by itself (ADS-B airborne velocity, 1-kt resolution), independent of any interrogation. From 24 October 2026 these messages are stored with the raw series of each landing (of each departure from 27 October). From 30 October 2026 they are used by **detection v8**, which runs *in shadow* beside the detection described in this guide (v7) for about a week: the headwind

from the true airspeed of both Comm-B registers — BDS 5,0, and BDS 6,0 with its indicated airspeed converted to TAS (pressure altitude, temperature from the flight's own Mach and TAS) — against the ADS-B ground speed of 0.5 s before each reply. On the landings of 24–30 October it gave a headwind every 0.5 s instead of 2 s (2.3 times the values, slightly less noise) and resolved more of the short ramps of 400–600 m: 41 ramp advisories against 30 and one WSA. Its result is stored with every landing and departure and shown in the Approach History's hover (*v8 (shadow, no alert)*); it raises no alert. After the week the two are compared and v8 replaces v7 if it holds (details in the study log).

Headwind from one register. Within a single BDS 5,0 reply,

$$h = \mathrm{TAS} - \mathrm{GS} = W_x + \mathrm{TAS}\,(1 - \cos\beta), \tag{2}$$

where W_x is the headwind along the track and β the crab angle (heading minus track). No heading, magnetic declination or wind-vector calculation is involved, so none of their errors enter. Both speeds are rounded to 2 kt in the register; the resulting noise is about 1.1–1.2 kt per reply [17]. The crab term is about 2 kt at $\beta = 10$ and changes only slowly along an approach, so it cancels almost completely in a ramp.

The wind vector. With BDS 6,0 added, the full wind is the ground vector minus the air vector: wind = (GS, true track) − (TAS, true heading). The true heading is the magnetic heading plus the declination of the World Magnetic Model, corrected for each airframe's own heading offset, which MODE-S Wind learns continuously. The wind vectors are less precise than TAS − GS (a 1° heading error is about 2.5 kt of crosswind at 140 kt) but they can be compared between aircraft flying in different directions. Mode S derived winds of this kind have been validated against other observations in several studies [13].

Where the data end. The receiver near Tuusula sees the aircraft down to a few hundred feet: the lowest point followed is 476 ft above mean sea level on a median approach (about 300 ft above the runway), depending on the runway. The last part of the approach and the landing roll are not seen. Turboprops send no Mach, and a few aircraft do not answer BDS 5,0 at all; they are simply not analysed.

3 The detector step by step

Figure 1 shows the chain from the replies to an alert. The same code (`collector/shear.py`) runs live on every aircraft on final, and once more over the whole approach when the landing is recorded.

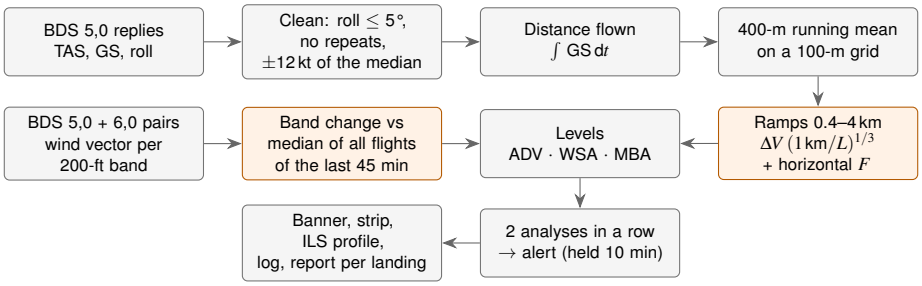


Figure 1. From the aircraft's replies to an alert. Top: the headwind ramp along the path of one aircraft (TAS − GS). Bottom: the change of the wind between flights (wind vectors). Both give a level; an event found in two analyses in a row becomes an alert.

3.1 Cleaning the replies

A reply is not used when the aircraft is banked more than 5° (in a turn the speeds and the track change quickly), when it repeats exactly the previous register content within 6 s (the transponder sent an old value again), or when TAS or GS are outside the plausible range. The remaining values are compared with their running median over ±7.5 s; a value more than 12 kt away from it is a corrupt reply and is dropped. The scatter of the accepted values about that median is the aircraft's own noise σ, used below for the significance test (1.24 kt on a median approach).

3.2 Distance along the path

A windshear ramp is a change over a *distance*, so the replies are placed along the path. The distance flown is the ground speed integrated over time,

$$x_k = x_{k-1} + \tfrac{1}{2}\,(\mathrm{GS}_{k-1} + \mathrm{GS}_k)\,(t_k - t_{k-1}), \tag{3}$$

not computed from the reported positions: positions freeze when GNSS is jammed, which happens regularly in this region, while the ground speed of BDS 5,0 comes from the inertial system and keeps running. Height is the barometric altitude of the tracker (ft above mean sea level).

3.3 Smoothing

The headwinds are averaged over 400 m of path (a running mean, at least two replies in the window) on a 100-m grid. This removes the reply noise and single gusts, which last only a few seconds, while keeping changes that extend over 400 m or more — the scales on which windshear affects a transport aircraft. Changes over less than about 400 m are deliberately smoothed out.

3.4 Headwind ramps

For every pair of grid points L apart, with $L = 0.4$, 0.6, 0.8, 1, 1.5, 2, 3 and 4 km, the detector takes the change of the smoothed headwind in the direction of flight,

$$\Delta V = \bar{h}(x+L) - \bar{h}(x) \qquad (\Delta V < 0\text{: loss, } \Delta V > 0\text{: gain}), \tag{4}$$

and scales it to a **1-km equivalent**

$$V_{1\,\mathrm{km}} = |\Delta V| \cdot \left(\frac{1\ \mathrm{km}}{L}\right)^{1/3}. \tag{5}$$

The scaling is that of the Hong Kong Observatory's glide-path windshear alerting (Section 4): the hazard of a ramp grows with ΔV and falls only slowly with its length. A 15 kt change over 1 km is therefore as significant as 12 kt over 0.5 km, 19 kt over 2 km or 24 kt over 4 km (Fig. 3). The ordinary increase of the wind with height — often 10–15 kt between 1 800 ft and the ground, spread over 8–10 km of approach — stays well below the levels, whereas the same change concentrated in a kilometre or two does not.

A ramp is accepted only when

- the profile has data on at least 60 % of its length;
- the change is **significant**: $|\Delta V| \geq 5\,\sigma\sqrt{1/n_1 + 1/n_2}$, where n_1 and n_2 are the numbers of replies in the two 400-m windows at its ends;
- for a short ramp ($L \leq 600$ m): there are at least three replies at each end, and the headwind does not come back by more than half of the change within the next 400 m — so a single odd stretch of replies cannot raise it.

Only ramps in the **low-level zone** count for alerts: one end below 1 600 ft above the threshold, or within 5 NM of it. (The ICAO layer is 1 600 ft; the 5 NM lets the detector see the final part of the approach where the aircraft are still above 1 600 ft.) For each approach the strongest loss and the strongest gain are kept.

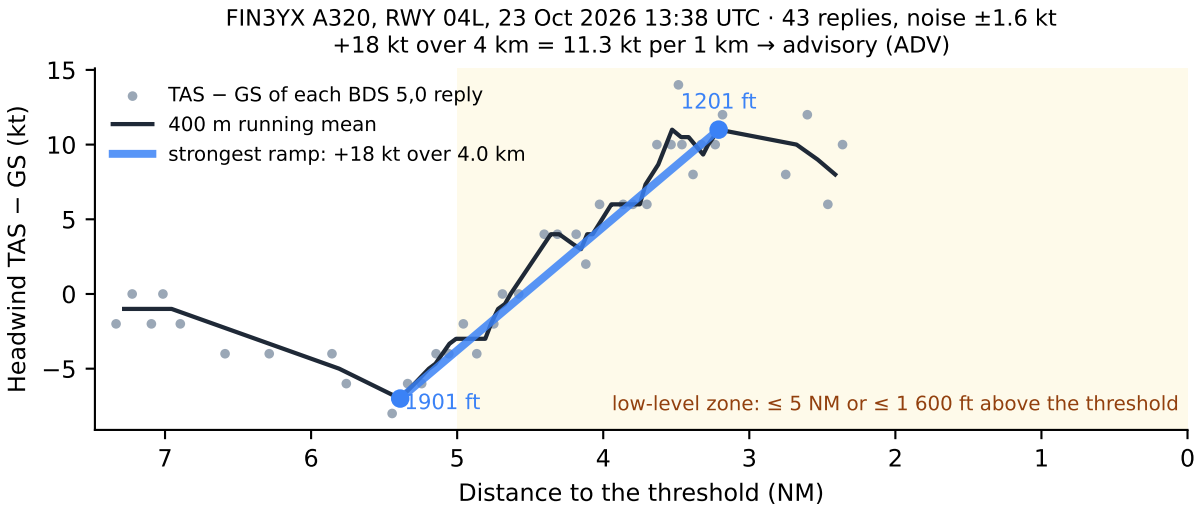


Figure 2. A headwind ramp. Grey dots: TAS − GS of each accepted reply of an A320 approaching runway 04L on 23 October 2026 in a shear layer behind a front; black: the 400-m running mean. Between 5.4 and 3.2 NM from the threshold (1 900 → 1 200 ft) the headwind increased by 18 kt over 4 km, which is 11.3 kt per 1 km: an advisory. Every landing on runways 04L/R in that afternoon showed the same layer (Section 6).

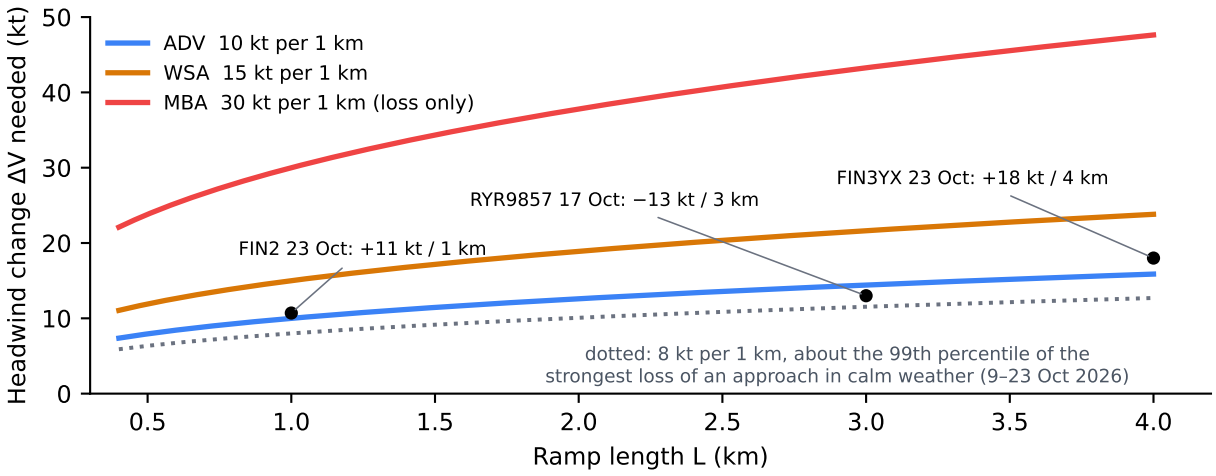


Figure 3. The headwind change a ramp needs to reach each level, as a function of its length. The dotted line is the 99th percentile of the strongest 1-km-equivalent loss of the approaches of 9–23 October 2026, a period of calm weather [17].

3.5 Horizontal F-factor

The same profile gives the horizontal part of the F-factor over 1 km of path: the largest headwind loss over 1 km divided by the time it took to fly it and by g,

$$F_h = -\frac{\bar{h}(x+1\,\mathrm{km}) - \bar{h}(x)}{g\,\Delta t}. \tag{6}$$

The vertical part ($-w/V$) needs the vertical wind, which cannot be observed from Mode S data, so F_h is a lower bound of the real F. On ordinary approaches at EFHK it is around 0.01 (99 % below 0.03); it is shown in the alert details and the landing reports but does not set the level. As a reference, airborne windshear systems are designed to warn at a full F of about 0.1–0.13 [5].

3.6 Change against the earlier flights (band comparison)

A single aircraft cannot see a change *in time*: when a front passes, each approach on its own can look smooth while the wind at 1 000 ft has turned completely between two landings. The second part of the detector therefore compares each flight with the flights before it.

For each 200-ft band from 400 to 1 800 ft above mean sea level, the aircraft's wind vector $\mathbf{u}_b$ (the mean of its wind pairs in the band, heading-calibrated) is compared with the **median** wind $\tilde{\mathbf{u}}_b$ of **all landings and departures of the previous 45 minutes, whatever their runway** (at least three of them; the median of the east and north components, so one odd flight does not move it). The difference is taken as a headwind along the aircraft's own direction ψ:

$$a_b = -(\mathbf{u}_b - \tilde{\mathbf{u}}_b)\cdot\hat{\mathbf{e}}_\psi, \qquad \hat{\mathbf{e}}_\psi = (\sin\psi, \cos\psi), \tag{7}$$

so $a_b = -25$ kt means 25 kt *less* headwind than the flights before. The band with the largest $|a_b|$ is reported. Bands with a mean wind above 80 kt or a spread of the wind pairs above 15 kt are left out (corrupt replies); a landing's band needs at least two wind pairs.

Why wind vectors and not TAS − GS here: TAS − GS is the headwind along the flight's own track, so it can only be compared with flights in the same direction. At Helsinki-Vantaa the usual configuration lands on one runway and departs from another (e.g. land 15, depart 22R), and 55 % of the departures had no landing in their own direction within 45 minutes. With wind vectors every flight is a reference for every other: the median approach is compared with 12 earlier flights, and 97 % of the landings have a band comparison.

Landings with landings, departures with departures (from 27 October 2026, report version 5). Against the MEPS model the band winds of the landings read on average 1.45 kt *less* and those of the departures 1.38 kt *more* headwind along their own direction (600–1 800 ft, 9–26 October 2026, 9 439 and 6 789 bands, on every runway direction and largest in steady flight) — an airspeed effect of the flight phase, not a real wind difference. Mixing the two, a departure started about 3 kt high against the landings: 27 of the 36 departure advisories and alerts of 9–27 October were "more headwind". Each flight is now compared with the flights of its own phase when a band has at least five of them; otherwise with all flights, each reference and the flight itself first corrected by its phase's bias along its own direction ($\mathbf{u}_{\mathrm{true}} = \mathbf{u} + b\hat{\mathbf{e}}$, $b = -1.45$ kt for a landing, $+1.38$ kt for a departure). Replayed over 9–27 October: the mean departure change moved from $+1.6$ to -0.3 kt, the departure flags became balanced (19 more and 16 less headwind instead of 27

and 9; 35 instead of 36), the landings' levels did not change, and of the five departure advisories of 26–27 October two remained. The hover tells which references were used.

Turboprop departures: limits 13 / 19.5 kt (from 27 October 2026, report version 6). The climb-outs of the ATR 72 scatter more than those of the jets — at 1 200–1 800 ft the band change varies by 3.2–3.8 kt (standard deviation) against 2.7–2.9 kt for jet departures, where the ATR accelerates, retracts its flaps and makes its first turn — and with the same limits 2.9 % of the turboprop departures got a band advisory or alert, against 0.6 % of the jets. For the departure of a turboprop (ATR, Dash 8, Saab, Fokker 50, Dornier 328, Jetstream, Beech 1900 / King Air, PC-12, Caravan, L 410, Brasilia, Metro) the band limits are therefore 1.3 times higher: ADV from 13 kt and WSA from 19.5 kt. Jets and all landings keep 10 / 15 kt. A small, steady offset of some types (up to about 1.4 kt, e.g. E190 and A220 slightly more, A330 slightly less headwind) was also found, but it hardly changes the alerts and is not corrected. Replayed over 11–27 October: 9 turboprop departures flagged instead of 21; jets and landings unchanged. The hover of such an alert says *Turboprop climb-out: limits 13 / 19.5 kt.*

Departures. Helsinki-Vantaa works in waves; during a departure wave there may be no landing for 20–30 minutes. Each climb-out is therefore compared in the same way, and becomes a reference for the next 45 minutes. Its runway is found from the first position and the initial climb track (track within 25° of a runway direction, first position within 1 NM of that runway's extended centreline); climb-outs after a go-around are not used. A departure climbs four to five times as steeply as a 3° approach, so a band may rest on a single wind pair; such a band gets a level only when a neighbouring band (±200 ft) shows a change of at least 10 kt in the same direction. Departures give band alerts only, no ramps (the acceleration on the climb-out makes TAS − GS harder to interpret).

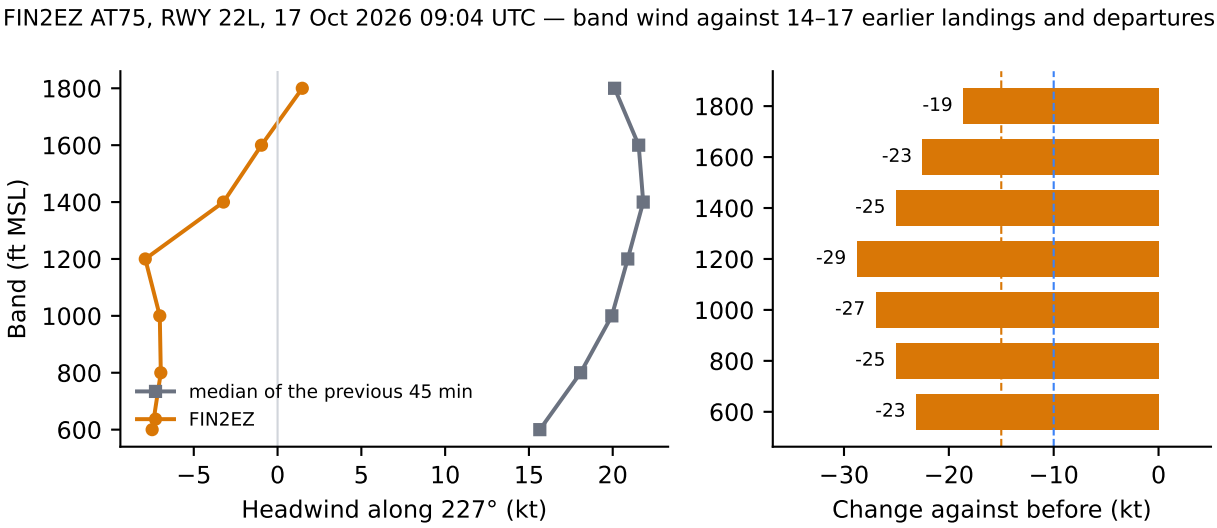


Figure 4. The band comparison of FIN2EZ on 17 October 2026. Left: the headwind along runway 22L in each band, FIN2EZ (orange) and the median of the 14–17 flights of the previous 45 minutes (grey). Right: the difference. At 1 200 ft the wind was 334°/27 kt instead of about 220°/21 kt: 29 kt less headwind than before — a windshear alert (WSA), although the approach itself was smooth (no ramp above the noise).

3.7 Levels

The names follow the runway-oriented alerts of the US airport windshear systems (LLWAS and TDWR), where a *wind shear alert* is a headwind loss of 15–29 kt or a gain of 15 kt or more, and a *microburst alert* a loss of 30 kt or more [9]. MODE-S Wind adds an advisory level below them.

Table 1. Alert levels.

Level	Badge	Ramp (1-km equivalent)	Band change	Colour
WS advisory	ADV	≥ 12 kt, loss or gain; 10–12 kt only when it stands out from the other flights' profile (below)	≥ 10 kt (headwind along the flight); turboprop departure ≥ 13 kt	blue
Windshear alert	WSA	≥ 15 kt, loss or gain, **and** a headwind change $\lvert\Delta V\rvert \geq 14$ kt	≥ 15 kt; turboprop departure ≥ 19.5 kt	amber
Microburst alert	MBA	≥ 30 kt, **loss** only (and $\lvert\Delta V\rvert \geq 14$ kt)	—	red

A real change of 14 kt (from 28 October 2026, report version 7). A ramp becomes a windshear or microburst alert

only when its headwind change itself is at least 14 kt. The Hong Kong Observatory's lidar alerting does the same: it uses $\Delta V/L^{1/3}$ to choose the most severe ramp, but alerts on the change [6, 8]. The 1-km equivalent lifts a short ramp — a 400-m ramp by a factor of 1.36 — so a change of only 11 kt, measured by three or four replies at each end, could reach 15 kt per km; such a ramp now stays an advisory. From 1 km up the rule changes nothing, because there the 1-km equivalent is never larger than the change itself. Replayed over the 4 155 stored landings of 9–28 October 2026, the ramp windshear alerts went from 3 to 1 (two 11-kt changes over 400 m on 26 October became advisories), and the advisories from 83 to 85. The comparison with the Hong Kong method is in the windshear study log (Section 10 and Phase 6).

Small ramps against the normal profile (from 26 October 2026, report version 4). The wind usually increases with height, and in a stable evening boundary layer it does so steadily all the way down the approach. On 25 October 2026 at 20:00–22:30 UTC, with a south-easterly wind of 15 kt at the ground, the landings on runway 15 met a median headwind of 20 kt at 400 ft and 35 kt at 1 800 ft; 10 of the 39 landings had a ramp of 10–14 kt per km and became advisories, while the comparison between flights saw nothing unusual. That is the normal profile of the evening, not windshear. A ramp of 10–12 kt (1-km equivalent) is therefore an advisory only when it is also at least 10 kt beyond the change the other flights of the last 45 minutes had over the same heights — their median band winds (landings and departures of any runway, as for the band comparison), turned into headwind along this runway. From 12 kt, or when there are no reference flights (a quiet night), the ramp alone decides as before; windshear and microburst alerts are never made relative. The report keeps the reference change and the ramp beyond it with each ramp, and the live analysis on final uses the same rule. Over 9–26 October (3 672 landings) this left 26 of the 59 ramp advisories; of the 21 advisories of 24–26 October, 13 remained — 11 with ramps of 12–14 kt and 2 that stood out from the profile by 11–12 kt.

Confirmed or isolated (from 27 October 2026). A shear layer or a front is met by every aircraft that passes it; a turbulent eddy in a strong wind over land by one. Each ramp alert is therefore compared with the other landings on the same runway direction within ±15 minutes: it is **confirmed** when at least one of them had a ramp of the same sign (loss or gain) of 7 kt or more per 1 km whose middle lies within ±300 ft of this one's, and **isolated** when none had. This is a label only — the level is not changed, because the first aircraft into a new shear is always alone at that moment. The live alert can only be compared with the landings before it; the stored report is checked again when the 15 minutes after the landing have passed, and the alert log is updated. On 26 October 2026, a turbulent day in a strong southerly flow, the shear layer at 1 150–1 350 ft on 22L at 13:08 UTC was confirmed by four landings, while seven of the twelve ramp alerts of the day had no confirming neighbour (Section 6).

3.8 From an event to an alert

The live analysis runs whenever new replies of an aircraft have arrived — about every 3 s, with each sweep of the tracker. Each analysis gives its *events*: a ramp loss, a ramp gain or a band change at some level. An event becomes an **alert** when it is found in **two analyses in a row**, so a single bad reply cannot raise one. The alert stays active for 10 minutes after it was last seen and is then removed. If it escalates (advisory → WSA) it is logged again; a stronger value at the same level updates the log entry. The server keeps the alert log of the last 24 hours (at most 100 entries) in memory, so it is shown also when the page was closed, and writes a short line to its log file for every new alert.

3.9 A windshear report for every flight

When a landing is recorded, the whole analysis is run once more over the complete final approach and stored with the landing: the replies used and their noise, the strongest loss and gain ramps with their heights and distances, the horizontal F-factor, the band comparison and the overall level. Each departure gets its band comparison, and from 27 October 2026 also its headwind ramps along the climb-out as a **trial**: the departure's replies are now kept, the same ramp analysis is run on them (the low-level zone being the first 1 600 ft above the airport or 5 NM from it) and the result is stored with the level it *would* have — but it raises **no alert** until enough climb-outs have been checked (the acceleration and flap retraction in the first 1 500 ft may affect TAS − GS, and a fast climb gives only 15–25 replies in the zone). These reports are shown in the Approach History (Section 5.5). When the server starts, older landings that still have their raw replies (kept 90 days) and the stored departures get their report in the background, in time order, so that each flight is compared with the flights before it.

4 The scientific basis

The ramp method. The Hong Kong Observatory (HKO) alerts windshear at Hong Kong International Airport, where terrain-disrupted airflow causes frequent windshear, with a Doppler lidar that scans along the glide paths. Its glide-path algorithm (GLYGA) builds a headwind profile along each approach and departure path and looks for *ramps*: headwind changes over lengths of 400 m to 4 km, ranked by $\Delta V/L^{1/3}$ [6, 3]. The form of this criterion comes from earlier analyses of aircraft flight-data recorders, in which windshear encounters were identified from the headwind along the flight path [6, 3] (the methods of Woodfield and Woods, 1983, and Jones and Haynes, 1984, as cited there); HKO describes the severity of a ramp as primarily a function of $\Delta V/L^{1/3}$ [8]. Chan, Hon and Shin added short-scale gradients to the ramps

and caught 5–10 % more of the pilot-reported windshear [7]; Hon and Chan showed that removing "gentle ramps" — long, smooth headwind changes that exceed the threshold only because of their length — cuts false alerts [8].

MODE-S Wind applies the same idea to a different sensor. Instead of a lidar profile along the glide path at one instant, it uses the profile measured by each aircraft itself as it flies down the path. Two differences follow: the aircraft sees the air along its own path only (not the whole path at once), and its profile is available only after it has flown through the air. In return, every equipped aircraft is a sensor, also in rain and fog where a lidar is blind, and no equipment at the airport is needed. The ramp lengths (0.4–4 km) are those of GLYGA; the 1-km normalisation, the 15 kt alert level, the significance test and the persistence test of short ramps are choices of MODE-S Wind, fitted to the noise of the Mode S data.

The hazard index. The F-factor and its 1-km averaging length come from the NASA/FAA windshear programme [4, 5]. The horizontal part can be computed from the same profile; the downdraft part cannot, which is why F_h is reported for information and does not set the level.

The alert levels follow the ground-based systems: LLWAS and TDWR alert on the runway-oriented headwind change, with the 15 kt and 30 kt (loss) limits of Table 1 [9], and 15 kt is the common definition of significant windshear [2, 3]. The extra condition of a 14-kt change (version 7) follows the HKO lidar alerting, whose threshold is 14 kt [6, 8].

The comparison between flights has no direct precedent that I know of. It is closest to the idea of using aircraft reports as objective windshear observations ("automated pilot reports"), which HKO pursues with flight data from airline quick-access recorders [10], and to LLWAS, which compares the wind at different places at the same time; here the wind at the same place is compared at different times.

Mode S data for windshear. Wind and temperature derived from Mode S Enhanced Surveillance have been studied mainly for weather forecasting [13]. A restudy of the MODE-S Wind data in October 2026 found no published work that detects low-level windshear from passively received Mode S replies; the Hong Kong Observatory has noted that the low latency and high sampling rate of Mode S winds make them suitable in principle for real-time windshear alerting [17]. The restudy also measured the noise floor of TAS − GS and replayed 3 084 landings, which led to the design described here and to the retirement of the six earlier browser algorithms (their description is kept in the guide *Windshear detection v1 — retired algorithms*).

Table 2. MODE-S Wind among windshear sensors (simplified).

Sensor	Measures	Strength	Limitation
LLWAS (anemometers)	surface wind around the runways	continuous, direct	surface only, fixed points
Doppler radar (TDWR)	radial wind in precipitation	microbursts in rain	weak in dry air
Doppler lidar (HKO)	radial wind in clear air, along the glide path	whole path at once	blind in heavy rain and fog
Airborne (reactive / predictive)	the aircraft's own F-factor / radar ahead	on board, certified	one aircraft, at the moment
MODE-S Wind	headwind along each aircraft's path, wind per band	every aircraft, any weather, no airport equipment	after the fact, only where aircraft fly, no downdraft

5 The Windshear page

The Windshear page follows the aircraft on final approach and shows the results of the detection. Figure 5 shows it during the wind change of 17 October 2026 (Section 6).



Figure 5. The Windshear page, reconstructed for 17 October 2026 09:05 UTC from the stored data of that morning (the live alert state is not stored, so it was rebuilt from the detector's results; the Approach History, the statistics and the map are the real page on the stored database). (1) alert banner, (2) flight strips with the WS badge, (3) map with the approach centrelines, (4) Approach History in WS mode, (5) Wind Rose / ACI panel, (6) ILS profile with the alert bands, (7) Windshear Alert log with the alert-level selector and the engine status line, (8) Today's statistics, (9) METAR and TAF.

5.1 Banner, flight strips and ILS profile

The **alert-level selector** in the header of the Windshear Alert panel (Banner off / ADV ≥10 kt / WSA ≥15 kt / MBA ≥30 kt; default WSA, remembered in the browser) sets the lowest level that is shown in three places:

- the **alert banner** at the top of the page, one entry per active alert, led by its message (Section 5.3), e.g. `22LA WSA 29K– 3MF · at 1200 ft vs the last 45 min [FIN2EZ]` for a band change, or `04LA ADV 18K+ 5MF · over 4.0 km, 1901→1201 ft [FIN3YX]` for a ramp; the heading reads `WINDSHEAR ADVISORY`, `WINDSHEAR ALERT` or `MICROBURST ALERT` after the highest level shown, followed by the `RESEARCH` marker. Pointing at an entry gives the spoken form. (Until 28 October 2026 the entry began `WSA · RWY 22L · 29 kt ▼LOSS.`);
- the **WS badge** on the aircraft's flight strip (Fig. 6): ADV blue, WSA amber and MBA red, the last two pulsing; pointing at the badge gives the message;
- a coloured band on the **ILS profile** over the height range of the alert (the ramp from its start to its end height, or the 200-ft band of a band change), labelled with the message, e.g. `22LA WSA 29K– 3MF` (until 28 October 2026 `WSA 29kt▼`).



Figure 6. Banner and flight strips. RYR9857 on 22L at 3.4 NM carries a WSA badge; its strip shows the wind 333°/26 kt, i.e. a 7 kt tailwind for runway 22L, where the flights 20 minutes earlier had a headwind of about 20 kt. FIN2AT, behind it, has no alert yet.

A landed aircraft (from 29 October 2026). After touchdown the receiver still hears an aircraft for a while, and it stayed on the page for about one and a half minutes at its last position, about 0.6 NM before the threshold (in 600 landings of October 2026). Now an aircraft within 2 NM of the threshold and below 1 000 ft whose position has not changed for 15 seconds is shown faded — a grey square with `landed` on the map, a grey dot on the ILS profile and a dimmed strip with

`LANDED?` — and after 30 seconds it is no longer shown; the automatic barb selection moves on to the next aircraft. Only the display changes: the landing is recorded and analysed as before. The mobile view `/app` does the same.

5.2 The alert log

The **Windshear Alert** log lists the alerts and the go-arounds of the last 24 hours, newest first, one line each: time (hh:mm), the message (Section 5.3) and the callsign, e.g. `09:04 22LA WSA 29K- 3MF FIN2EZ`. Until 28 October 2026 the line read time, level, runway (`dep` for a departure), change with ▼ loss or ▲ gain, the ramp length (`1.2km`) or the band (`@1200ft`) and the callsign, as in Figs. 8 and 9. A ramp alert ends with its confirmation: ✓2 when two nearby landings saw the same ramp, `iso` when none did, `iso?` while the landings after it are not yet known. The log always shows all levels, whatever the selector says. Hovering an entry gives the details (Figs. 8 and 10): the time to the second, the message with its spoken form, and for a band change the reference headwind and the number of earlier flights, for a ramp the heights and distances of its ends, the 1-km equivalent, the horizontal F-factor and the confirmation by the nearby landings (Fig. 9). Clear hides the entries in this browser only. The description of the six retired algorithms of detection v1 is no longer linked here (from 30 October 2026); it remains in the *User guides* tab as *Windshear detection v1 — retired algorithms (legacy)*.

From 29 October 2026 an alert that came during a **low-level jet** or a **surface inversion** (Section 5.8) ends with a small tag `LLJ`, `INV` or `LLJ+INV`, and its details add a line such as *Boundary layer then: low-level jet 1750 ft 27 kt from 134°* (Fig. 7). Descending through a jet gives the kind of headwind change that a ramp alert measures, so the tag points to a possible cause; whether the jet explains the alert depends on its direction against the runway.

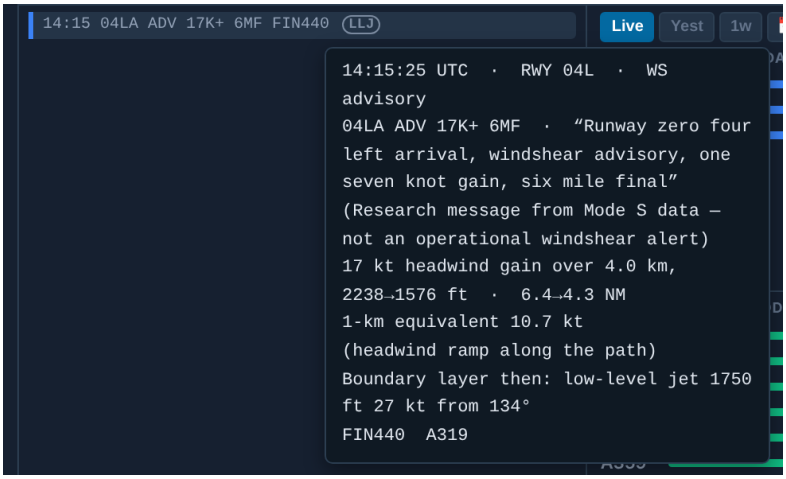


Figure 7. The advisory of FIN440 on 23 October 2026, rebuilt from its stored report: a headwind gain of 17 kt over 4.0 km between 2 238 and 1 576 ft on final 04L. A south-easterly low-level jet of 27 kt had its nose at 1 750 ft in that hour, hence the tag `LLJ`. The jet blew mostly across 04L, so it explains the gain only in part — the tag says the jet was there, not that it caused the alert.

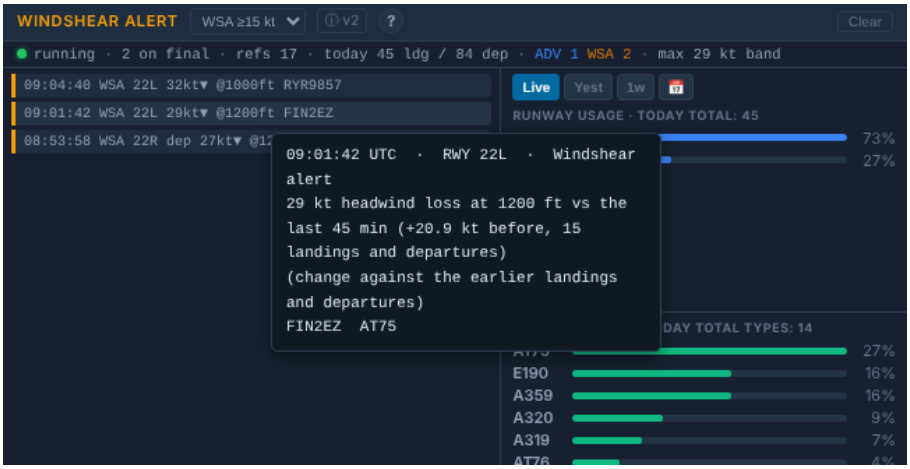


Figure 8. The alert log of 17 October with the details of the FIN2EZ alert: 29 kt headwind loss at 1 200 ft, against +20.9 kt in the 15 landings and departures of the previous 45 minutes. The first entry of the morning is the departure FIN863 at 08:54 (`22R dep`).



Figure 9. The ramp alerts of 26–27 October 2026 with their confirmation (rebuilt from the stored reports): the shear layer of FIN6KB on 22L seen by four landings (✓ 4), the short ramps of FIN9EJ, FIN9LK and others in the turbulent southerly flow by none (`iso`).

5.3 The alert message

From 28 October 2026 every alert also has a short message in the form that airport windshear systems give to the controllers — the US Low-Level Windshear Alert System (LLWAS) and Terminal Doppler Weather Radar (TDWR), and the Hong Kong Observatory's Windshear and Turbulence Warning System. The controller reads it out to the pilot almost word for word, so the format is familiar to both:

Part	Example	Meaning
Runway + phase	`15A`, `22RD`	the runway, `A` arrival (on final) or `D` departure (climb-out)
Level	`WSA`	`WSA` windshear alert, `MBA` microburst alert; `ADV` advisory is our own level, not used by those systems
Change	`15K+`, `29K−`	the headwind change in knots, − loss, + gain (a ramp: its ΔV; a band: the change against the last 45 minutes)
Where	`RWY`, `2MF`, `1400FT`	where it is first met: `RWY` on the runway (under 0.5 NM), `2MF` two-mile final; a climb-out gives the altitude instead

So `15A WSA 15K+ 2MF` is read *"Runway one five arrival, windshear alert, one five knot gain, two mile final"*, and `22RD WSA 15K− 1400FT` *"Runway two two right departure, windshear alert, one five knot loss, at one thousand four hundred feet"*. Numbers are spoken digit by digit, nine as *niner*, as in radiotelephony.

For a ramp on final the distance is that of its upper end, where the aircraft met it first; without a position, and for a band change on final, it follows from the height above the threshold on the 3° glide path (318 ft per NM, so the band at 1 200 ft MSL on 22L is `3MF`). A departure gives the altitude of the band, rounded to 100 ft, because the band comparison has no position along the climb-out. The message is made on the server and kept with the alert; it is shown in the banner, the alert log, the ILS profile and the flight-strip badge, on the phone in the ILS approach view (Fig. 10), and in the server's log line. The detailed values — ramp length, heights, the 1-km equivalent, the reference flights — stay in the hovers and in the windshear reports.

Research, not operations. The message borrows the operational format only so that it is easy to read. It is always marked `research`: it comes from passively received Mode S replies of aircraft that have *already* flown through the air, and it is not a windshear warning (Section 7).

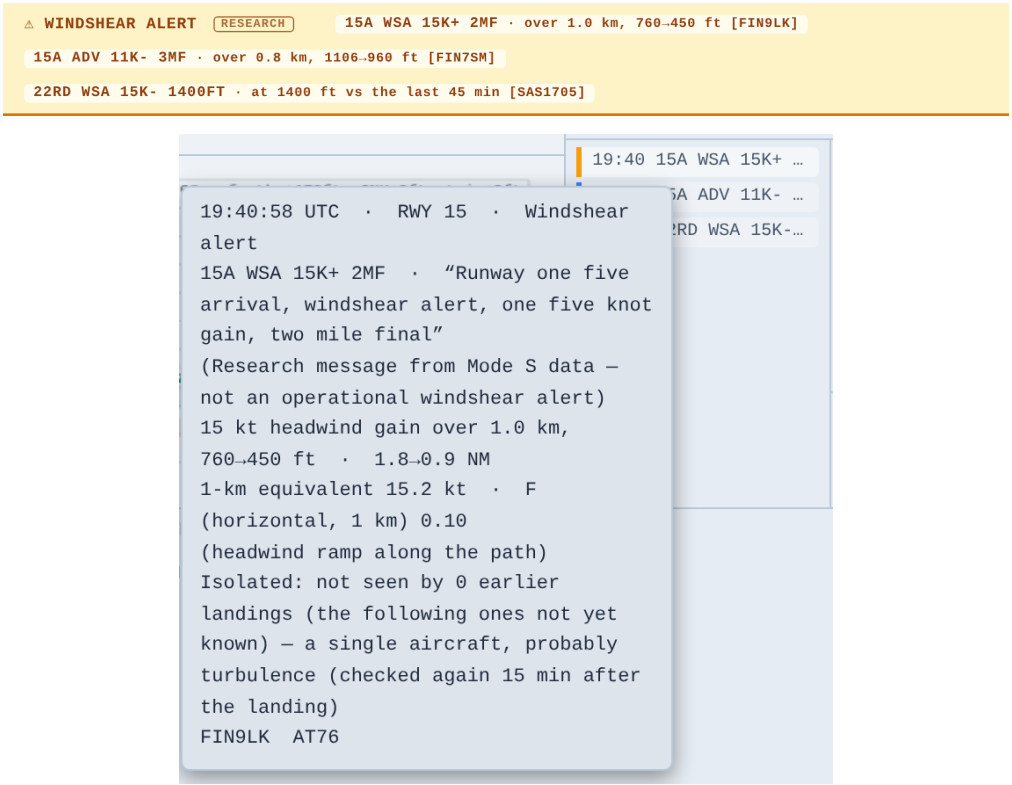


Figure 10. The messages in the light theme: the banner with three active alerts (two landings on runway 15, a departure from 22R) and the RESEARCH marker, and the alert log with the hover of the first entry — the message, its spoken form and the details. Example alerts in a test set-up, not a real event.

5.4 The engine status line

Under the header of the Windshear Alert panel one line tells whether the detection on the server is working (Fig. 11), e.g. • `running · 2 on final · refs 17 · today 45 ldg / 84 dep · ADV 1 WSA 2 · max 29 kt band`:

- the **dot**: green *running* when the last analysis is recent; amber *no analysis for … min* when aircraft are in the ILS corridors but nothing has been analysed for more than two minutes — then no alert does *not* mean calm air, and the server log is worth a look; grey *waiting for traffic* when nothing has been analysed since the server started;
- the aircraft **analysed now**: landings on final and departures (`dep`) analysed in the last 30 s;
- `refs`: the landings and departures of the last 45 minutes that the band comparison can use. With fewer than three it turns grey with *band check waiting* — normal at night, when only the ramps can raise an alert;
- **today**: the landings and departures that got a windshear report since midnight UTC, and how many of them reached each level — the engine is working even when it finds nothing;
- **max**: the strongest value of the day, the larger of the strongest ramp (1-km equivalent) and the largest band change. In calm weather it stays around 5–10 kt; when it climbs towards 10–15 kt the day is getting interesting.

When the panel is narrow the aircraft analysed now and the strongest value are left out. The hover gives all details: the time of the last analysis, the aircraft analysed and in the corridors, the reference flights, today's counts, the strongest ramp and band change with callsign and time, the median reply noise and lowest height followed of today's landings, the last alert and how long the server has been running. The day's numbers are rebuilt from the stored reports when the server starts, so a restart does not reset them.

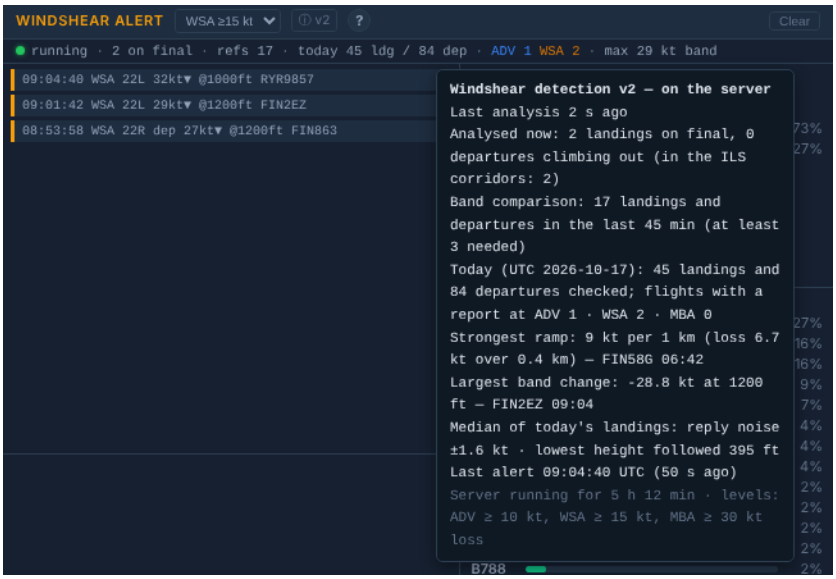


Figure 11. The status line with its details, 17 October 2026 09:05 UTC (reconstructed as Fig. 5; the day's numbers are those of the stored reports): 45 landings and 84 departures checked since midnight, one advisory and two alerts (FIN863 and FIN2EZ), 17 reference flights, and the largest band change of the day, −28.8 kt at 1 200 ft.

5.5 Approach History: the WS view

The Apch Hist button opens the Approach History, one row per landing (Landings) or, from 27 October 2026, per departure (Departures, at the right of the title; the choice is remembered). For the departures the band columns show the wind of the climb-out along its runway direction, and the WS view its band comparison and its *trial* ramps in italics, e.g. *(ADV trial)* — what the ramps would give if they alerted. The ACI is computed for landings only. Its display selector has seven views: the wind per height band (Wind, HW, XW, HW+XW), the Approach Conditions Index (ACI, Section 5.6), the windshear report (WS, Fig. 12) and, from 29 October 2026, the turbulence (EDR, Section 5.7). The WS view has seven columns:

Column	Content
Level	ADV / WSA / MBA of the landing, — for none; after it the confirmation of a ramp alert (✓n / `iso`)
Loss, Gain	the strongest ramp each way, as `7 kt/3.0` = 7 kt over 3.0 km; coloured by its level when in the low-level zone
F 1km	horizontal F-factor over 1 km
Band Δ	the largest change of a band against the previous 45 minutes, e.g. `-29 @1200`; coloured by its level
Low	the lowest height followed (ft MSL)
n	replies used, with the noise (± kt)

Hovering a row shows the whole report. For FIN2EZ (Fig. 12) it reads: *Windshear report — Windshear alert · Strongest loss: none above the noise · Band 1200 ft: −28.8 kt against 20.9 kt in the last 45 min (15 landings and departures) · Followed … 40 replies, noise ±1.3 kt.* Landings without stored raw replies (contact lost early, or before 24 October 2026 without a series) show —. From 29 October 2026 a flight with a level during a low-level jet or a surface inversion gets the `LLJ` / `INV` tag beside it, and the hover ends with the boundary layer of that hour (Section 5.8).



Figure 12. Approach History, WS view, 17 October 2026. The landings on 22L until 08:54 had small band changes (−3 to −7 kt) and a few 3–4 km ramps of 6–9 kt (the normal increase of the wind with height, well below the advisory level); FIN2EZ at 09:04 had −29 kt at 1 200 ft.

5.6 The other panels

ILS profile and wind barbs. The profile shows every aircraft on final against the 3° glide path (corrected for the QNH and the threshold elevation), with its trail. Barbs draws the winds measured by one aircraft (a clicked strip, or Auto: the lowest one) along its approach, HW / XW add the head- and crosswind components, and the METAR wind is drawn at the threshold. The barbs are collected on the server for every aircraft on approach (every 400 ft or 0.5 NM, Hi: every 150 ft or 0.2 NM), so a newly opened page shows the whole approach so far. Comparing the barbs of successive aircraft is the quickest way to see a ramp or a change between flights for yourself.

The aircraft on the map (ATC style from the evening of 29 October 2026; Fig. 13). Each aircraft is an open square framed in the colour of its wind source — larger on the approach, smaller for the other traffic; the one whose barbs the profile shows has a thin white frame (near-black in the light theme) and is a little smaller, so it is only lightly marked (from 30 October 2026). Its label, in a regular-weight radar-label font with the callsign and the altitude in the same size (from 30 October 2026), gives the callsign and under it the altitude as a controller's label shows it: up to the transition altitude of 5 000 ft `A` and the hundreds of feet on the METAR QNH (`A023` = 2 300 ft), above it the flight level (`FL061`), with ↑ or ↓ when the aircraft climbs or descends 300 fpm or more. Traffic to the right-hand runways (04R, 22R) is labelled on the left of its square, so that the labels of the two parallel finals do not overlap. After a **go-around** the label shows a flashing red GA after the callsign (from 30 October 2026), from the moment the go-around is detected until the aircraft turns onto its next approach — or until it disappears from the map as a climb-out.

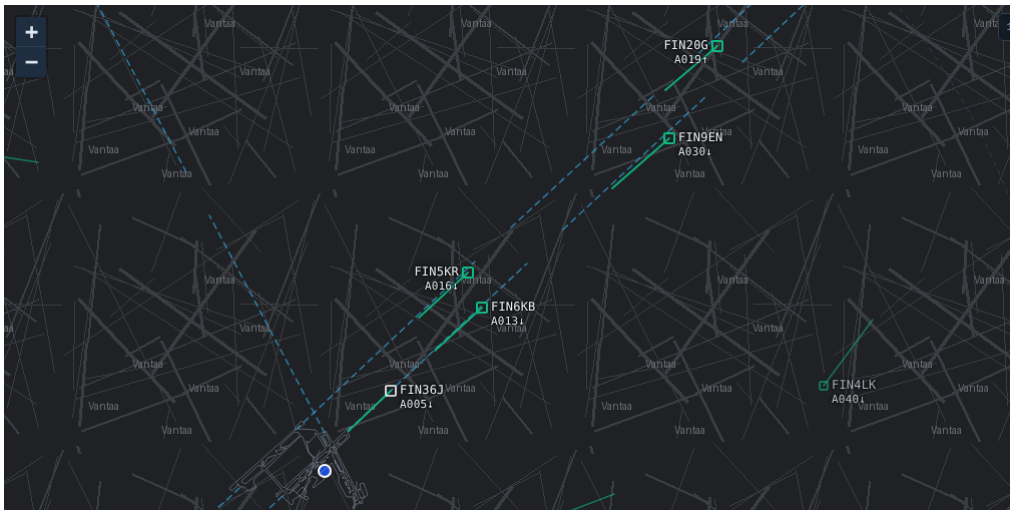


Figure 13. The map with the approach traffic of 27 October 2026, 12:58 UTC (replayed in a test, without a METAR, so the altitudes are on standard pressure): five aircraft on the finals 22L and 22R — the 22R traffic (FIN5KR, FIN20G) labelled on the left; the lowest, FIN36J at 500 ft, with the white frame of the barb aircraft; FIN20G climbing (`A019↑`); FIN4LK, not on an approach, dimmer. The lines ahead of the squares are the 30-s speed vectors.

Speed vectors on the map (from 29 October 2026; 30 s from the start since the evening of 29 October). The Vec button above the map steps through Off, 30 s, 1 min and 2 min: each aircraft then gets a line to where it will be after that time at its ground speed along its track, as on a radar screen, in the colour of its square. A faster aircraft has a longer line; at 140–250 kt below 5 000 ft, 30 s is 1.2–2.1 NM and 1 minute 2.3–4.2 NM, about the spacing on final. A landed aircraft gets none. The choice is remembered in the browser.

Label levels (from 30 October 2026). The Lbl button next to Vec steps through five levels of the aircraft labels on the map: CS — the callsign only; Type — the ICAO aircraft type only (e.g. AT75); CS+Alt — the callsign over the altitude, as before (the default); +Type — the type after the callsign (FIN456 AT75) over the altitude; Full — in addition the altitude selected on the autopilot panel (MCP / FCU) after the altitude, signed with the vertical rate, e.g. A023+020 (at 2 300 ft, climbing or level, 2 000 ft selected) or A045-030 (descending, 3 000 ft selected). The selected altitude is shown only when the aircraft has sent it within the last 2 minutes; otherwise the climb / descent arrow stays. With the type or the selected altitude the labels are a little smaller. The level is remembered in the browser.

Map styles, history plots and range rings (from 30 October 2026; Fig. 14). Dark and Grey show the street map, dark or light (they follow the page theme), with the symbols in the colour of the wind source as described above. ATC — the same radar-display style as on the Live Map — replaces the street map with a plain background, dark grey in the dark page theme and light grey in the light one, with the airport layout, the ILS (short dashes) and RNP (long dashes) finals and the coastline. The symbols are monochrome: light below 6 000 ft, grey above, the aircraft whose barbs the profile shows yellow (blue in the light theme). The labels have up to three lines and a leader line from the square, and they are moved automatically so that they do not overlap each other, the other squares, the history plots and speed vectors or the panels over the map; a label keeps its place while that stays free. The lines are the callsign, the level with the trend and the selected level in hundreds of feet (`019↑060`: at 1 900 ft, climbing, 6 000 ft selected) and the ground speed in tens of knots with the type (`18 AT75`); the level is the altitude on the METAR QNH up to 5 000 ft and the flight level above, without the letters. The Lbl levels choose how much: CS or Type one line, CS+Alt two, +Type three and Full three with the selected level; a go-around shows the flashing red GA after the callsign. The Rings button, shown only in the ATC style, adds range rings from the airport every 5 or 10 NM out to 30 NM (off at the start). A legend at the bottom left gives the colours and marks the display as research — not for air traffic control: the positions are those received by this station, not certified surveillance data. The style follows the conventions of European radar displays such as Thales TopSky, whose screens and settings are not public; it is not a copy of any one system. It replaces the earlier flat ATC (light grey) and Black map styles.

The Hist button (in every style) adds **history plots**: small filled squares at the aircraft's earlier positions, one about every 6 s, the older ones fainter — Off, 5 (at the start), 10 or 20 plots, about 30 s to 2 minutes. The style, the plots and the rings are remembered in the browser.

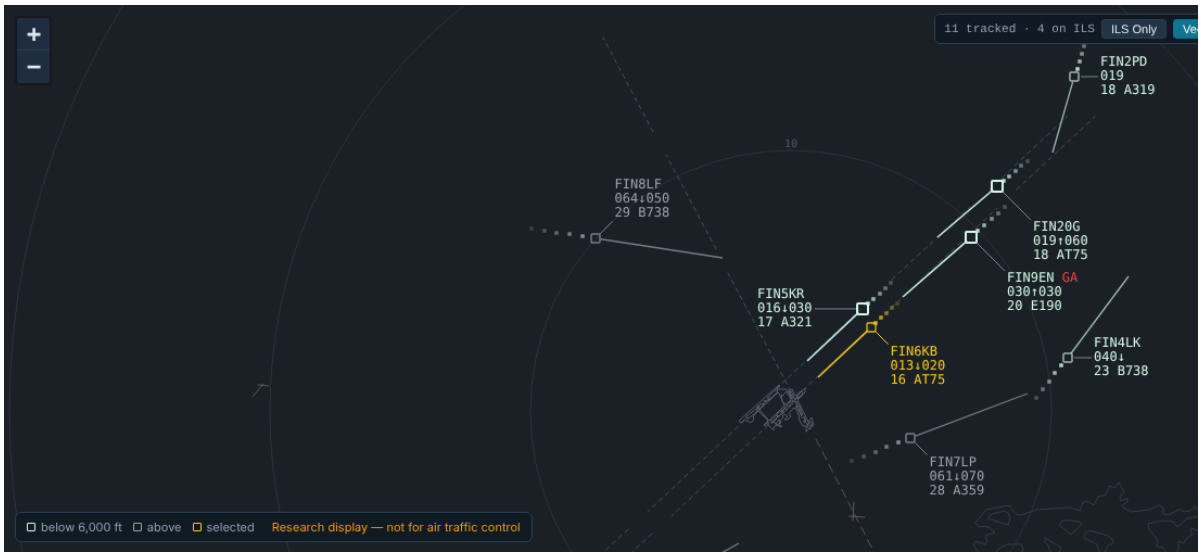


Figure 14. The ATC map style in the dark page theme with Hist 5, Vec 1 min, Lbl Full and Rings 10. A test: the approach traffic of 27 October 2026, 12:58 UTC, replayed and moved along its tracks to draw the history plots; the selected altitudes and the go-around of FIN9EN are test values, not observed. FIN6KB is the aircraft whose barbs the profile shows.

Hover (ATC style, from 30 October 2026; Fig. 15). The Hover button, shown in the ATC style after Rings, switches on an *extended label*: when the mouse is on an aircraft's square, its label grows into six lines with what the receiver has just heard from the aircraft, on a dark backing (light in the light page theme) with a thin frame, placed first so that the other labels move out of its way. It stays open while the mouse is on the square or on the label itself. The lines are:

- the callsign, the type and the registration;
- the level with the trend, the selected level (`sel 060`; `sel ---` when none was heard within 2 minutes), the vertical rate in ft/min and `AP` or `APP` when the aircraft's ADS-B reports the autopilot or the approach mode engaged;
- the ground speed, the true airspeed (BDS 5,0), the indicated airspeed and the Mach number (BDS 6,0);
- the magnetic heading (BDS 6,0), the magnetic track (the true track less the local magnetic declination), the drift (the track less the heading, `L` or `R`) and the bank angle (BDS 5,0, `ROLL L2` = 2° left wing down);
- the wind, its headwind (`HW`) or tailwind (`TW`) and crosswind component on the track (`XW 18R` = 18 kt from the right) and the static air temperature (`SAT +4`, °C);
- the distance and bearing from the airport, the NACp and the squawk when heard.

The values are fetched when the mouse arrives and every 2 seconds while it stays — read from the receiver's memory, nothing is stored — and a value not heard within 20 seconds shows as `---`: an aircraft that the radars are not interrogating for these Comm-B registers has no TAS, IAS, Mach, heading or roll. The symbol's tooltip is not shown while Hover is on. Hover is off at the start and remembered in the browser; the Live Map has the same button. Label dragging by hand is not provided: the labels are re-placed at every update.

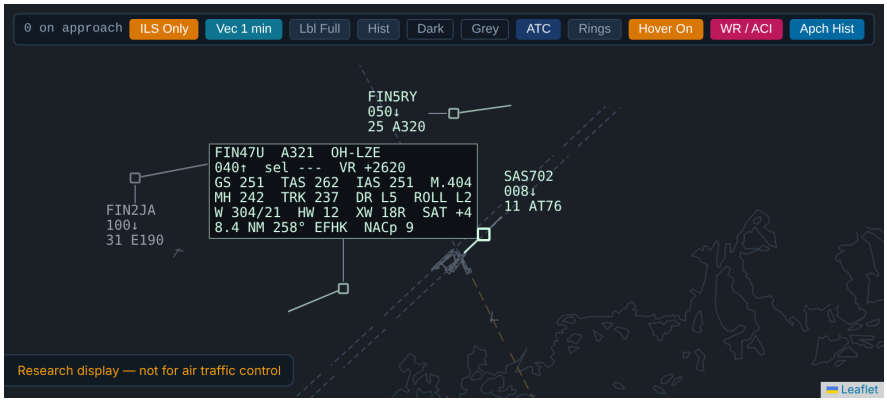


Figure 15. Hover On: the mouse on FIN47U, whose label has grown into the extended label (dark page theme, Lbl Full, Vec 1 min). A test: four aircraft of 30 October 2026, 08:29:50 UTC, from the stored observations — positions, levels, speeds and FIN47U's BDS 5,0 / 6,0 values as stored — drawn on the page with the METAR QNH 1018 of that time. No selected altitude was stored, hence `sel ---`.

How the page opens. On the first visit (from 27 October 2026) the page opens ready for watching the approaches: Barbs, Hi, Auto and HW on, the Approach History open and the ACI chart over 12 hours. Every change you make — the barb buttons, the Approach History open or closed, the WR / ACI / WS view and its range — is remembered in the browser, and the page opens the same way next time. From 30 October 2026 this includes the map's position and zoom (and its style, Section 5.6); the home button (a small house) beside the zoom buttons brings the map back to the airport at the default zoom.

Wind Rose. Compares the METAR surface wind (cyan) with the mean wind of the approaches of the last 30 minutes below 2 000 ft (green); Hist shows how the low-level wind direction has drifted over the last 3 or 6 hours.

Approach Conditions Index (ACI). The ACI button of the same panel plots one dot per landing: an index from 0 (smooth) to 10 (very rough) of how much the aircraft was moved on final (Fig. 16). From 30 October 2026 the turbulence (EDR, Section 5.7) of the same landings and of the departures is drawn below it on the same time axis, so that the two can be compared: the ACI is how much the aircraft moved, the EDR how rough the air was. It is built from the quick fluctuations of airspeed (45 %), vertical rate (30 %) and bank angle (25 %) — each the RMS about a ±7.5 s running median, so the slow, planned changes of an approach are removed — scaled between a calm and a rough reference, averaged over the segments 3 000–1 000 and 1 000–200 ft, and divided by a class factor (turboprops 1.5, business jets 1.25, regional jets 1.15, narrow- and widebodies 1.0) because light aircraft are moved more by the same air. Levels: below 2 *Smooth*, 2–4 *Light*, 4–6 *Choppy*, 6–8 *Rough*, from 8 *Very rough*. The ACI is not part of the windshear detection and not a turbulence report: it measures the aircraft's response (including the autopilot or the pilot), but it follows the surface wind well (median about 1 at 0–8 kt, 3 at 9–12 kt, 4.3 at 13–16 kt) and is a useful companion to the windshear alerts on a gusty day. It is still provisional.

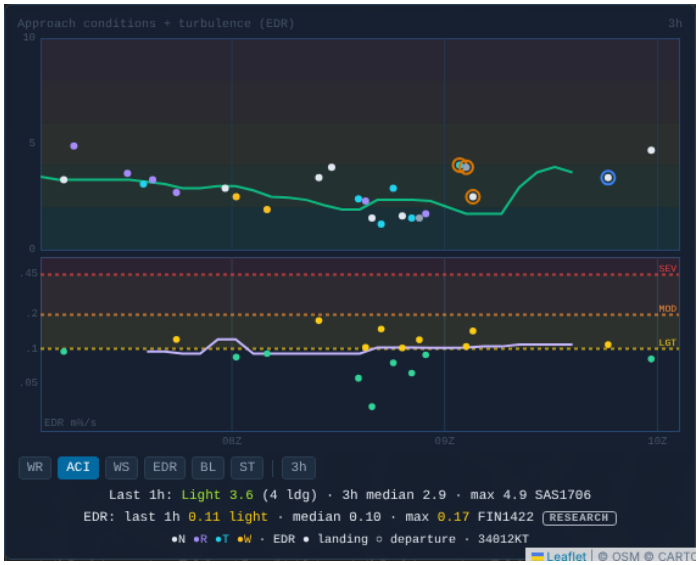


Figure 16. The ACI view of the Wind Rose panel, 17 October 2026 07:05–10:05 UTC: above one dot per landing, coloured by aircraft class, with the running median (green; rings: the landings with a windshear advisory or alert); below the EDR of the same landings (dots) and of the departures (rings), mostly near the light line 0.10. The wind change did not make the approaches rough: windshear and turbulence are different things.

Windshear (WS view). The third button of the panel, WS, shows the *windshear strip* in knots of headwind change over the chart range (Fig. 17; from 30 October 2026 alone, the full height of the panel — until then under the ACI chart). In the strip each **landing** has its strongest headwind **ramp** along its own path, scaled to 1 km (Section 3), as a stem from zero — **loss down, gain up** — coloured by its level when it lies in the low-level zone, and its **band change** against all flights of the last 45 minutes as a filled diamond; each **departure** has its band change as a hollow diamond; **go-arounds** are triangles on the time axis with a dotted line through the chart. Dashed lines mark ±10 kt (ADV), ±15 kt (WSA) and −30 kt (MBA, losses only); the scale is ±20 kt, widened to ±40 or ±50 kt on a strong day. Hovering any mark gives the flight's windshear report and its ACI, and the readout adds the alerts of the chart range. In the ACI view, too, a landing with an advisory or alert has a ring around its dot in the level's colour.

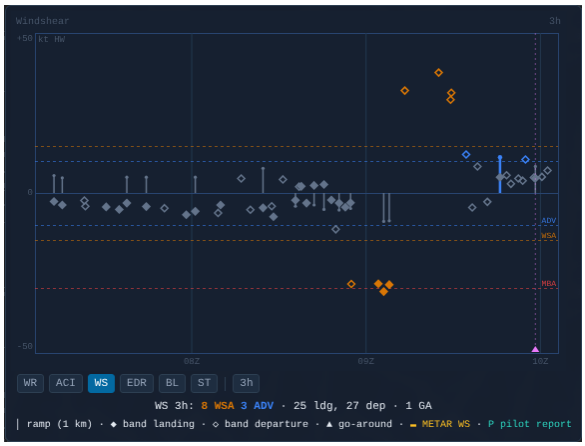


Figure 17. The WS view, 17 October 2026 07:05–10:05 UTC: the band losses of −29 to −32 kt at 1 000–1 200 ft of FIN2EZ, RYR9857 and FIN2AT on 22L at 09:04–09:07 (orange filled diamonds; the departure FIN863 had −26 kt at 08:54), and from 09:13 the departures on runway 04 with the reversed wind, +20 to +38 kt (hollow diamonds, WSA and ADV); the small grey stems are ordinary ramps of 6–9 kt per 1 km. The triangle at 09:58 is a go-around (NSZ2845, 04R).

Space–time diagram (ST view, from 27 October 2026). The fourth button, ST, shows how the wind changes travel through the successive flights (Fig. 18). For every landing and departure of the chart range (the same 3h–1d button) the headwind change of *every* 200-ft band from 400 to 1 800 ft against the flights of the previous 45 minutes is computed as in the band comparison (Section 3.6) — the detection itself keeps only each flight's largest change. Time runs to the right and height upwards; each cell is the **median** of the flights in a 5–30-minute slot: **blue** = less headwind than before, **red** = more, from 3 kt to full colour at 15 kt; smaller changes are the normal scatter and stay background. A dot marks a cell with a flight at or beyond its advisory limit, a triangle above the plot a flight with a windshear level. **How to read it:** a front or a shear layer is a coloured streak through successive flights, often through several bands, and slopes when it descends; turbulence gives scattered single cells. Hover a cell for its flights, their changes and winds.

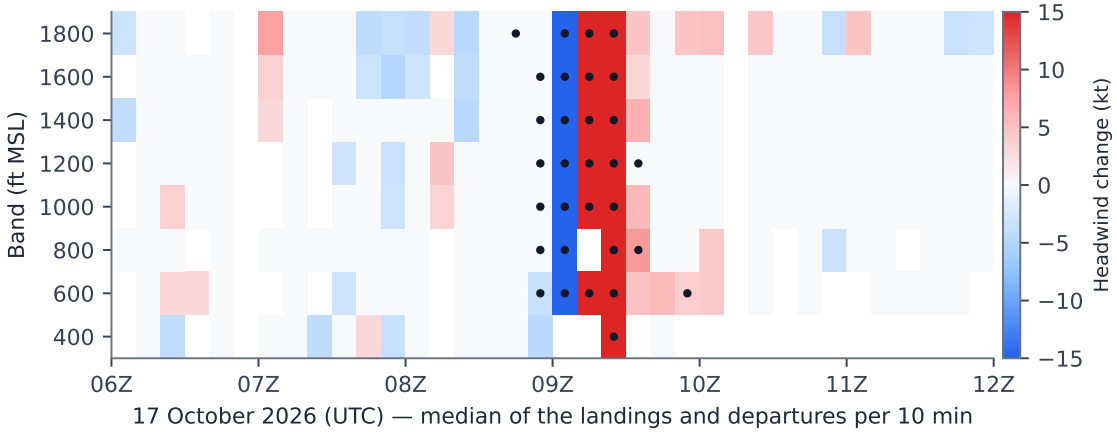


Figure 18. The ST view's picture for 17 October 2026 06–12 UTC (drawn here from the same data): the wind change at 09:00 is a blue streak through every band from 600 to 1 800 ft, followed by the red reversal of the departures after the runway change.

Go-arounds. The tracker detects a go-around from a sustained climb on final (or a climb after crossing the threshold) and logs it with the alerts (· GA 22L); a returning aircraft gets a 2nd APP badge. Every go-around is also stored with the windshear level active at that moment, so the log shows it for 24 hours — also when the page is opened later or after a restart of the server — and the go-arounds can later be compared with the alerts. A go-around shortly after a windshear alert is worth a closer look.

Today's statistics show the runway use, the go-arounds and the aircraft types of today, yesterday or the last week. An aircraft type with a landing whose windshear report has a level (ADV, WSA or MBA, the WS column of the Approach history) in that period is written in the colour of its highest level, with a small badge per level (e.g. WSA ADV×5); hovering the row lists those landings, newest first, with the time, callsign, registration, runway and the headwind ramp or band change that raised the level (from 30 October 2026). In the same way a type with a landing that came after a go-around gets a red outlined GA badge (GA×2: two go-arounds), its name red when it has no windshear level, and the hover lists those landings with the number of go-arounds before each; the **METAR / TAF** strip shows the latest

reports. A METAR wind that does not change while the alerts come is normal: the surface wind is measured at 10 m, the windshear is often several hundred feet higher.

5.7 Turbulence (EDR)

From 29 October 2026 MODE-S Wind also estimates **how turbulent** the air was for every landing and departure — research, not part of the windshear detection. The measure is the one ICAO uses, the **EDR** (eddy dissipation rate, $\varepsilon^{1/3}$ in $\mathrm{m}^{2/3}\,\mathrm{s}^{-1}$): ICAO Annex 3 [1] grades the *peak* EDR of a medium-size transport aircraft as light from 0.10, moderate from 0.20 and severe from 0.45. It is worked out from the same headwind TAS − GS as the windshear ramps, after the method of Kopeć et al. [15]:

1. the replies from 200 to 3 000 ft above the threshold are cut into **straight pieces** of about 2 km (track change at most 15°) — in a turn the headwind part of the mean wind changes and would look like turbulence;
2. each piece is detrended, and the mean squared headwind difference of all reply pairs 100–200 m apart is taken (the *structure function*), minus the noise of the 2-kt resolution of TAS and GS ($0.35\,\mathrm{m}^2\,\mathrm{s}^{-2}$, the calmest day);
3. for turbulence this grows as $2\,\varepsilon^{2/3} r^{2/3}$, which gives EDR at $r = 141$ m.

Each flight gets the EDR over the whole approach or climb, below and above 1 000 ft, and its strongest piece. In the study of 9–29 October 2026 (4 255 landings) the EDR followed the ACI (daily medians correlate at 0.98) and the surface wind (median 0.03 below 5 kt, 0.10 at 10–14 kt, 0.16 at 15–19 kt), was larger below 1 000 ft than above, and was nearly the same for all aircraft classes — it describes the air, not the aircraft, unlike the ACI. About 85 % of the landings get a value. Departures cross 200–3 000 ft in about 80 seconds and turn early, so only about a third of them had two straight pieces there; from 29 October 2026 their replies are therefore stored up to 5 000 ft and their EDR uses 200–5 000 ft (*hi* = 1 000–5 000 ft). Working out the wind vector from the heading, or using the turning pieces as well, was tried on the stored departures, but those values did not agree with the straight-piece ones (correlation 0.28 and 0.33).

On the Wind Rose panel the button EDR shows **turbulence (EDR) + windshear** on one time axis (Fig. 19; from 30 October 2026 without the ACI chart, which made the view too crowded — the ACI and the EDR are compared in the ACI view): the EDR strip above, the windshear strip below. The EDR strip has a logarithmic scale from 0.02 to 0.6 with dashed lines at LGT 0.10, MOD 0.20 and SEV 0.45; each landing is a dot and each departure a ring, green (smooth), yellow (light), orange (moderate) or red (severe), with the running median in purple. Hovering a dot gives the flight, its EDR, the two layers and the strongest piece. In this view the panel reaches down towards the bottom of the map, so the strips have more room, and the read-out starts with `EDR: last 1h 0.14 light · median 0.12 · max 0.19`.

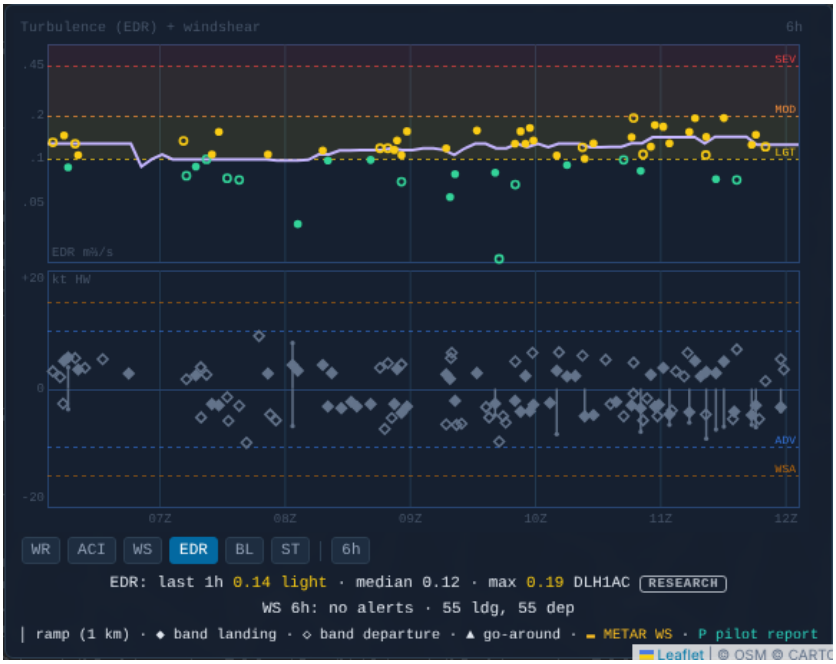


Figure 19. EDR view over 6 hours: the flights of 28 October 2026 06–12 UTC, replayed in a test. In a south-westerly wind the EDR of most landings was light (0.10–0.20), the departures (rings) a little lower; the windshear strip below shows no alerts — turbulence and windshear are different things.

In the Approach History the view EDR (from 29 October 2026, landings and departures; Fig. 20) lists each flight's EDR as a badge in its ICAO class colour — green smooth, yellow light, orange moderate, red severe — with Lo (200–1 000 ft), Hi (1 000–3 000 ft; departures to 5 000 ft), Peak (the roughest 2-km piece), n (pieces), the ACI for comparison and the

METAR wind. A filter beside the view (All, ≥ light, ≥ moderate) keeps only the bumpy flights; the title then reads e.g. `17 of 205 landings`. Values below the noise floor read `<0.01`; flights without straight pieces show —. Hovering a row gives the values, the ICAO limits and the boundary layer of the hour.

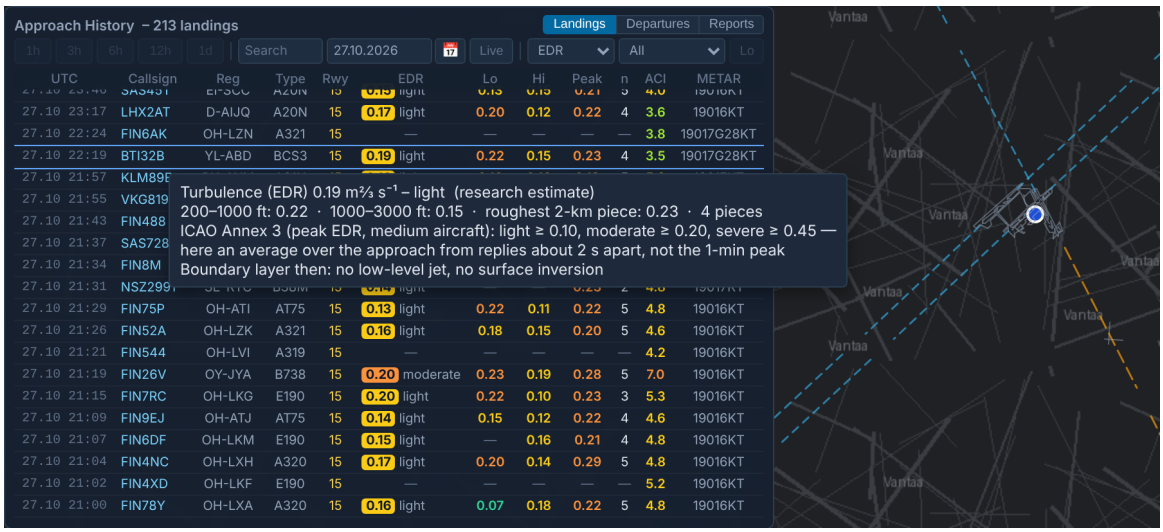


Figure 20. Approach History, EDR view, the turbulent 27 October 2026 (southerly 16–17 kt, gusts to 28 kt): most landings light (0.13–0.19), SAS728 and FIN26V moderate (0.20–0.21, roughest pieces 0.28–0.31) below 1 000 ft; the ACI follows. The hover of BTI32B shows the details.

A research value. The replies come about every 2 s, so only the 100–200 m scale is seen and the $r^{2/3}$ law is assumed there; the result is an average over the approach, not the ICAO one-minute peak, so the ICAO lines are a guide only. The values will be calibrated against the turbulence pilot reports (WXREP, ARS) that MODE-S Wind now stores.

5.8 The boundary layer: low-level jet and surface inversion

From 29 October 2026 MODE-S Wind also describes the lowest 5 000 ft at the airport hour by hour (research), because two boundary-layer features shape the wind on final:

- a **low-level jet** (LLJ) — a band of strong wind a few hundred metres up, typically on a clear night when the air near the ground has calmed down: the wind *increases* from the surface up to the jet's *nose* and *decreases* above it, so an aircraft on final meets a headwind change below the nose;
- a **surface inversion** — air above warmer than at the ground, on calm, clear nights: a stable layer in which turbulence dies out.

Every hour the server takes the median wind of all aircraft within 25 km of the airport in 250-ft steps up to 5 000 ft, and the temperatures of the landings and departures from 400 to 3 000 ft — and from 30 October 2026 of the departures up to 5 000 ft — against the METAR temperature. A **jet** is the strongest wind maximum at most 2 000 ft above the airport whose speed falls by at least 2.5 m/s and 25 % within 1 500 ft above it (after Baas et al. [14], made stricter); a **surface inversion** is the air at 600–1 000 ft at least 0.5 K warmer than the METAR temperature.

In the month to 29 October 2026 a jet was found in 10 % of the hours (16 % at night, 4 % at midday; typically at 1 500 ft and 23 kt) and a surface inversion in 15 % (23 % at night, 2 % by day; in half of the hours with a wind under 5 kt, never from 10 kt; up to +9 K on the clear, calm night of 19–20 October). In windy daytime hours the aircraft temperatures at 600–1 000 ft were 1.75 K below the METAR — exactly the cooling of well-mixed air over that height — so the aircraft read the low-level temperature without a bias. Under an inversion the turbulence (EDR) below 1 000 ft was nil.

The BL view. The button BL of the Wind Rose panel shows time against height, 0–5 000 ft, one column an hour (Fig. 21): the wind speed, or with the Wind / T button the temperature against the surface (red warmer — an inversion — blue colder); hatched where no aircraft flew. The jet's nose is a diamond with its speed, joined hour to hour (the jet core); a surface inversion is a red bar along the bottom. Hovering gives the wind of a cell, the jet (height, speed, direction, the drop above and the shear below) or the inversion. The read-out sums up the last hour and the range: `Last hour: low-level jet 750 ft 18 kt from 023° · surface inversion +2 K · 1d: jet 8 h · inversion 12 h.`

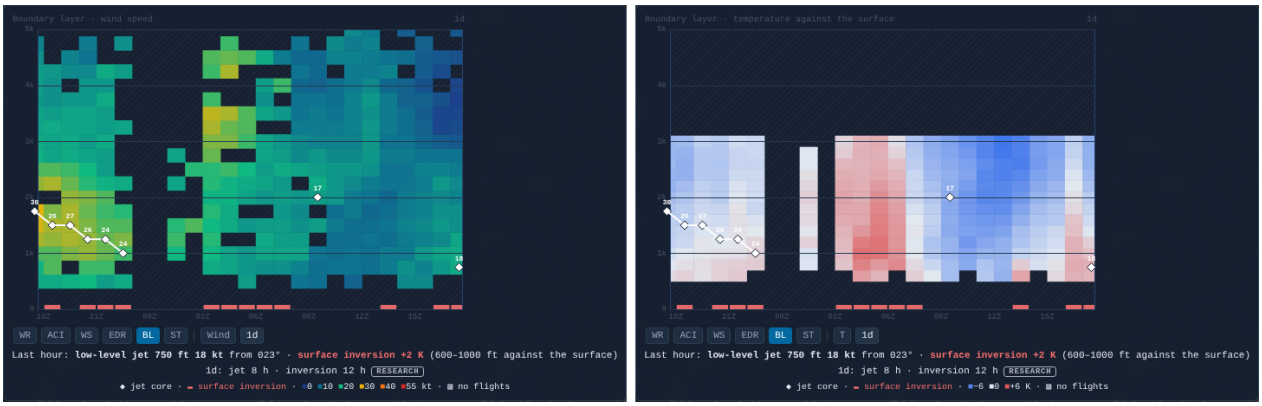


Figure 21. BL view, 17 October 2026 18 UTC – 18 October 17 UTC. Left, wind: in the evening a north-easterly jet of 30 kt sank from 1 800 to 1 000 ft and weakened to 24 kt; another formed in the afternoon of the 18th at 750 ft. Right, temperature: the night and morning (02–06 UTC) were warmer aloft than at the ground (red), the day colder (blue); red bars mark the hours with a surface inversion. Hatched: no flights (01–03 UTC).

The boundary layer is also shown where it helps to read the other products: as the `LLJ` / `INV` tag and the *Boundary layer then* line of the windshear alerts and reports (Section 5), on the Skew-T airport profile (a ◊ `LLJ` mark and a red `SFC INV` bracket; see the Skew-T guide), and as two fields of the Sounding page's time–height section.

5.9 METAR and pilot reports

From 29 October 2026 MODE-S Wind also keeps what was *reported officially*, so that the Mode S findings can be compared with it:

- **Every METAR of Helsinki-Vantaa**, with its windshear group. In Finland the group `WS R22L` (or `WS ALL RWY`; also written `WS RWY 22L`, sometimes after `RMK`) is given only in the *manual* METARs: windshear reported by aircraft or ATC during a landing or a take-off between the surface and 1 600 ft [16]. The automatic METARs of the night have none, and Finland publishes no SPECI — the METARs every 30 minutes are the whole official record. The history since 2018 can be loaded from the Iowa Environmental Mesonet archive; Helsinki-Vantaa has about eight days with a windshear group a year, in every season.
- **The pilot reports and SIGMETs** of the FMI aviation weather service (*ilmailusaa.fi*): the **ARS** (moderate or severe turbulence or icing, severe mountain wave, thunderstorm, volcanic ash) and the national **WXREP** — reports below the ARS criteria, such as *light* turbulence, light icing, light windshear, an *inversion* or the top of a CB [16]. A report within 30 NM of the airport (or naming it) and at or below 10 000 ft counts as **near**. The service keeps 24 hours; MODE-S Wind reads it every 30 minutes.

On the WS view (Fig. 22) a METAR with a windshear group is an amber bar along the top of the windshear strip, from the METAR to the next one, with the runway at its start; a pilot report near the airport is a teal **P** with a dotted line at the time it was observed. Hovering gives the whole message, and the readout adds e.g. `METAR WS R22L · 1 pilot report near`. When a pilot reports windshear on final, the landings around that time should show it too — or tell how far below their followed heights it was.

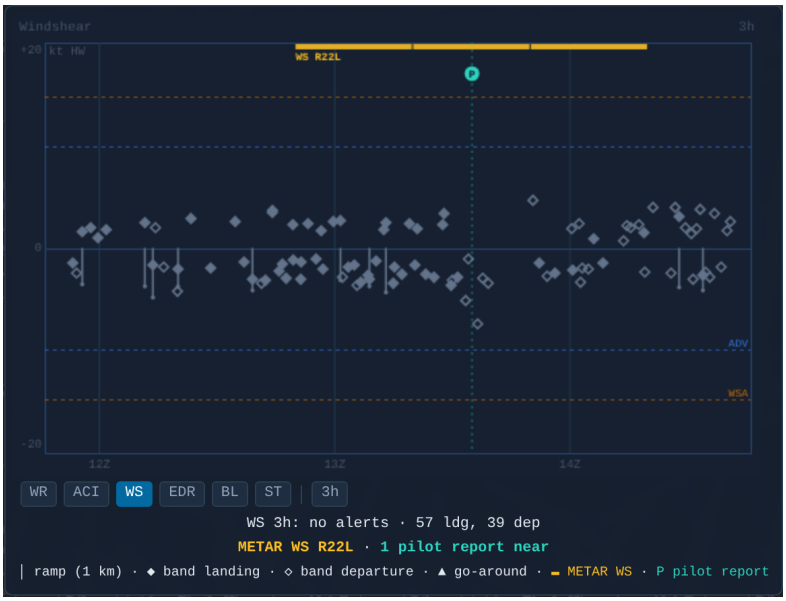


Figure 22. The WS view with the official reports — an example drawn with test messages (a METAR `WS R22L` for three half-hours and a WXREP `MOD WS SFC/800FT` on final 22L) over the landings and departures of an ordinary afternoon.

The messages themselves are in the Approach History: its third button Reports (title *METAR & pilot reports*) lists the METARs and the pilot reports of the chosen window or date, newest first (Fig. 23): time, kind (`METAR` with `AUTO` / `COR`, `ARS` red, `WXREP` teal, `SIGMET` violet; a teal dot = near) and the message, the windshear group in amber and the reported phenomena (`MOD WS`, `FBL TURB`, `INV` ...) in teal. The search box searches the text (`WS`, `R22L`, `TURB`, a callsign). In place of the display selector a filter: All, WS & near (the windshear METARs and the pilot reports near the airport) and WS archive (the same for every stored date). The phone has the same list in *History* (*List: METAR & reports*).

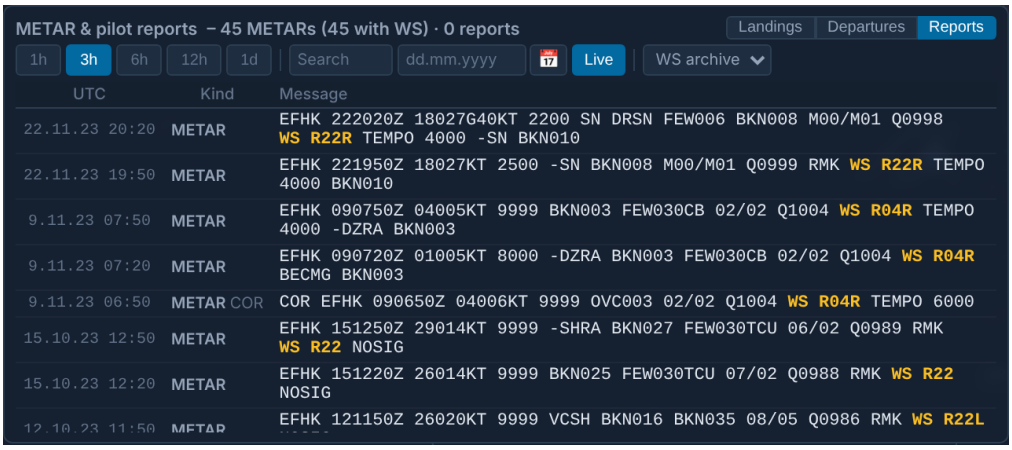


Figure 23. Reports with the filter WS archive: the METARs of Helsinki-Vantaa with a windshear group in 2023 (from the IEM archive), newest first — e.g. 22 November 2023 `18027G40KT ... SN DRSN ... WS R22R`, and 9 November in light wind and low cloud `WS R04R`.

6 Cases from October 2026

17 October 2026: a wind change at 1 000 ft. A cold front lay over southern Finland. Until 08:50 UTC the landings on 22L had a steady south-westerly headwind of about 20 kt at 1 000–1 200 ft. At 08:54 the departure FIN863 from 22R found 26 kt less headwind at 1 200 ft than the flights before it — a windshear alert, ten minutes before the next landing. From 09:04 the landings FIN2EZ, RYR9857 and FIN2AT each had 29–32 kt less headwind at 1 000–1 200 ft (Fig. 24); the wind there had turned from about 220° to 330°. The METAR hardly changed (220°/9 kt at 08:50, low cloud at 400 ft), and none of the approaches had a ramp above 10 kt per 1 km: each aircraft flew through a smooth profile, only a different one from its predecessors. From 09:13 the departures moved to runway 04 and saw the reversed wind (20–38 kt more headwind than the 22 departures before them). This is exactly the case the band comparison was made for.

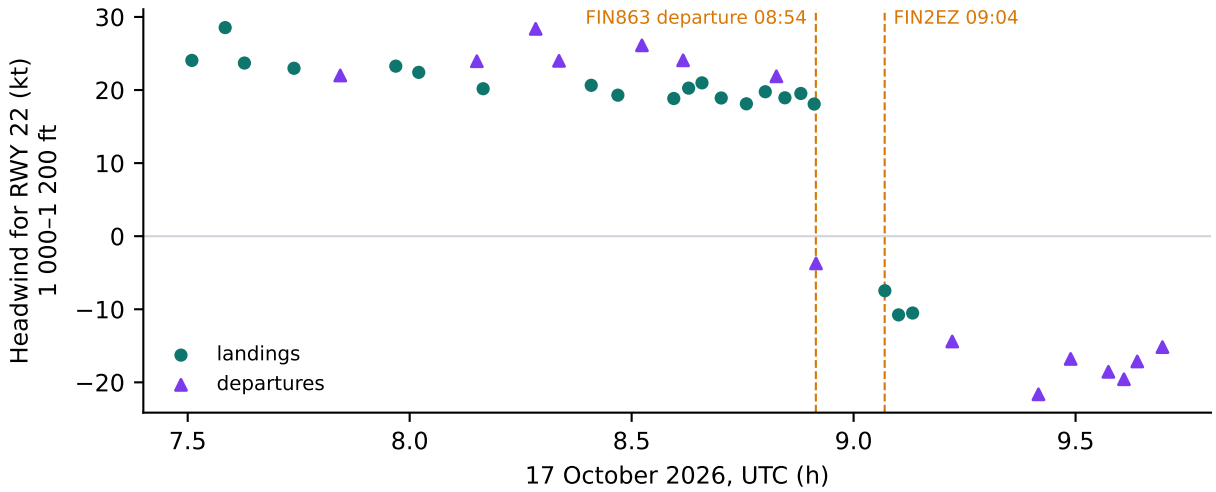


Figure 24. 17 October 2026: the headwind for runway 22 at 1 000–1 200 ft of every landing (green) and departure (violet) between 07:30 and 09:45 UTC, from their band wind vectors. The change arrived with the departure FIN863 at 08:54; the landings from 09:04 had a tailwind.

23 October 2026: a shear layer behind a front. On the afternoon of 23 October (11:30–14:15 UTC) every landing on 04L/R passed through a persistent shear layer at about 1 400–1 800 ft, with about 8 kt of tailwind above it and 10 kt of headwind below. Seven landings gave a ramp advisory (gains of 11–18 kt over 1–4 km, 1-km equivalent 10–11 kt; Fig. 2); none reached an alert, and the runway changes 04 → 15 → 22 during the frontal passage caused no false alarms.

26 October 2026: turbulence in a strong southerly flow. A warm, moist southerly flow (11 °C) blew 12–17 kt at the ground and 30–40 kt at 1 000–4 000 ft (Jokioinen soundings), with a well-mixed boundary layer. The approaches were the roughest of the month (ACI median 4.2, otherwise 1.1–2.2) and the replies on final scattered by 1.9 kt instead of the usual 1.2 kt. Twelve landings got a ramp alert (9 ADV, 3 WSA). One was a real shear layer: at 13:08 UTC every landing on 22L lost 7–10 kt of headwind between 1 350 and 1 150 ft, FIN6KB most sharply (14 kt per km). The 400-m ramp of FIN7SM (11 kt at about 1 100 ft) was seen by the landing three minutes earlier as well. The other alerts were single aircraft: short ramps at different heights each time, not repeated by the flights a few minutes before or after them, whose 200-ft band winds looked like their neighbours' — turbulent eddies that the aircraft really flew through, but not windshear in the operational sense. This day led to the confirmed / isolated label (Section 3.7).

Results so far. Table 3 summarises the stored reports of 9–24 October 2026. The weather was calm or moderate (the highest METAR wind 17 kt) and no windshear was reported at EFHK, so these numbers are mainly a false-alarm check; the levels still have to be confirmed in strong-wind and convective weather.

Table 3. Windshear reports 9–24 October 2026.

	Landings	Departures
Flights with a report	3 310	3 084
Ramp advisories (ADV)	38 (1.1 %)	—
Ramp alerts (WSA, MBA)	0	—
Band advisories (ADV)	0	21 (0.7 %)
Band alerts (WSA)	4 (0.12 %)	9 (0.3 %)
Noise of TAS − GS (median)	1.24 kt	
Lowest height followed (median)	476 ft MSL	
Horizontal F-factor over 1 km	median 0.011, max 0.047	
Earlier flights in the band comparison (median)	12	

Three of the four landing alerts and six of the nine departure alerts belong to the 17 October case (FIN863 and the five runway-04 departures of 09:13–09:34); two departure alerts came on 13 October and one at 04:42 UTC on 24 October during a night-time frontal passage, with 16 kt more headwind at 1 800 ft than the flights before.

7 Limitations and good practice

- **After the fact, and only where aircraft fly.** Every value describes air an aircraft has already flown through. An alert warns the *next* aircraft only if the shear persists — often for fronts and low-level jets, rarely for a short-lived

microburst. Between flights nothing is measured.

- **Not the whole approach.** The data usually end at 300–500 ft above the runway, where windshear is most dangerous; the last part of the approach is not seen.
- **Scale.** Changes over less than about 400 m are smoothed out by design; the typical size of the events that trigger on-board reactive warnings, a few hundred metres, is at the resolution limit of the data. The rate of the Comm-B replies is set by the WAM network's interrogation schedule; the ADS-B ground speed now stored may help here (Section 2).
- **No downdraft.** The vertical wind cannot be observed, so microburst downdrafts appear only through their horizontal outflow and the F-factor is a lower bound.
- **Slow changes.** The band comparison uses the last 45 minutes as its reference: a change that builds up gradually over an hour is absorbed by it.
- **Quality of the wind vectors.** The band winds rest on the heading calibration (about 3–4 kt of noise per band); a departure band often rests on one or two wind pairs.
- **Turbulence.** In a strong wind over land the short ramps (400–600 m) of single aircraft can reach the advisory or even the alert level; the confirmed / isolated label tells them apart from a shear met by several aircraft.
- **Landings and departures differ by about 3 kt.** An airspeed effect of the flight phase (Section 3.6). Since report version 5 each flight is compared with its own phase, or the bias is removed; the bias is an average and varies from day to day (4–9 kt between the phases in the strong wind of 26–27 October).
- **Aircraft types differ.** Turboprop departures scatter more than jet departures and have higher band limits (Section 3.6); a few jet types read 1–1.5 kt more or less headwind than the others, which is not corrected.
- **The levels are first values.** They were set on calm weather and one moderate front; they will be reviewed with strong-wind and convective cases.

Good practice. Read an alert together with its details (hover), the flights before and after it in the Approach History and the barbs on the ILS profile. Several aircraft in a row with the same change make it real; a single flight with a band change resting on one wind pair is a hint. Compare with the METAR (its `WS` group) and the pilot reports (Section 5.9), and in summer with the weather radar.

8 Quick reference

Parameter	Value
Headwind (ramps)	TAS − GS of each BDS 5,0 reply; dropped: roll $> 5°$, repeated register (< 6 s), > 12 kt off the ± 7.5 s median
Distance	ground speed integrated over time (not positions)
Smoothing	400 m running mean on a 100 m grid
Ramp lengths	0.4, 0.6, 0.8, 1, 1.5, 2, 3, 4 km; ≥ 60 % data coverage
Significance	$\lvert\Delta V\rvert \geq 5\sigma\sqrt{1/n_1 + 1/n_2}$; short ramps ($\leq 600$ m): ≥ 3 replies at each end and persistent over the next 400 m
Scaling	$V_{1\,\mathrm{km}} = \lvert\Delta V\rvert\,(1\,\mathrm{km}/L)^{1/3}$
Low-level zone	below 1 600 ft above the threshold or within 5 NM
Horizontal F-factor	largest headwind loss over 1 km / $(g\,\Delta t)$; information only
Band comparison	200-ft bands 400–1 800 ft MSL; median wind of the flights of the last 45 min (any runway, ≥ 3): the same phase when ≥ 5 in the band, else all with the phase bias removed (landing -1.45, departure $+1.38$ kt headwind); headwind change along the flight's direction
Band quality	landing ≥ 2 wind pairs; departure single-pair band needs a neighbour beyond the advisory limit the same way; bands > 80 kt or spread > 15 kt dropped
Departure runway	initial track within 25° of a runway, first position ≤ 1 NM from its centreline
Levels	ADV ≥ 10 kt (ramps 10–12 kt only ≥ 10 kt beyond the reference profile), WSA ≥ 15 kt (a ramp also $\lvert\Delta V\rvert \geq 14$ kt, from v7), MBA ≥ 30 kt loss (ramps only); band comparison of a turboprop's departure ADV ≥ 13, WSA ≥ 19.5 kt
Confirmation	ramp alert seen by another landing on the same runway direction within ± 15 min (same-sign ramp ≥ 7 kt per 1 km, middle within ± 300 ft); label only; final 17 min after the landing
Alerts	2 analyses in a row; held 10 min after last seen; log 24 h / 100 entries
Message	`15A WSA 15K+ 2MF`: runway + A/D, level, change in kt with − loss / + gain, `RWY` (< 0.5 NM) / `n`MF (upper ramp end, else height on the 3° path, 318 ft/NM) / departure altitude `1400FT`; spoken form in the hover; research only
Status line	amber when aircraft are in the corridors and nothing was analysed for 2 min; `refs` grey below 3; day = UTC
Departure ramps	trial only (from 27 Oct 2026): ramp analysis of the climb-out, zone first 1 600 ft above the airport or 5 NM; stored as `trial`, no alert
Space–time	ST view: headwind change of every band 400–1 800 ft of every flight vs the previous 45 min, median per 5–30 min, colour from 3 to 15 kt
Stored	`approach_history.shear_json`, `departure_profiles.shear_json` (report version 6, departures with `trial`); raw series `approach_series`, `departure_series`
Turbulence (EDR)	headwind TAS − GS 200–3 000 ft; straight 2-km pieces ($\leq 15°$); structure function at 100–200 m minus $0.35\,\mathrm{m^2 s^{-2}}$; EDR $= \sqrt{(D - N_0)/(2 \cdot 141^{2/3})}$; per flight all / below 1 000 ft / above / strongest piece; `edr_json`, `/api/windshear/edr-tracks`; research
Boundary layer	per hour: wind 250–5 000 ft (aircraft ≤ 25 km), T 400–5 000 ft vs METAR (above 3 000 ft departures only, from 30 Oct 2026); LLJ nose $\leq 2\,000$ ft above the airport, drop ≥ 2.5 m/s and 25 % within 1 500 ft; surface inversion 600–1 000 ft $\geq +0.5$ K; `bl_hours`, `/api/boundary`; research
Official reports	`metar_reports` (every METAR, windshear group `ws`; history: `tools/load_metar_history.py`, IEM), `pilot_reports` (ARS, WXREP, SIGMET of ilmailusaa.fi every 30 min; near = ≤ 30 NM and $\leq 10\,000$ ft); `/api/wx/reports`
API	`/api/windshear/state`: `ws_events`, `ws_log` (with `msg`, `say`), `ws_engine`; `/api/windshear/approach-history`: `shear`; `/api/windshear/departures?full=1`; `/api/windshear/spacetime`

References

[1] ICAO. *Annex 3 to the Convention on International Civil Aviation: Meteorological Service for International Air Navigation*. Montréal (wind shear warnings: runway level to 500 m / 1 600 ft).

[2] ICAO (2005). *Manual on Low-Level Wind Shear and Turbulence*, Doc 9817, 1st ed. Montréal.

[3] Lau, S.Y., Choy, B.L. (2005). Wind shear alerting at Hong Kong International Airport. WMO Workshop on Value Added Services for Air Navigation, Toulouse, 9–10 Sept 2005. Hong Kong Observatory reprint 604.

[4] Lewis, M.S., Robinson, P.A., Hinton, D.A., Bowles, R.L. (1994). *The relationship of an integral wind shear hazard to aircraft performance limitations*. NASA TM-109080, NASA Langley Research Center.

[5] Proctor, F.H., Hinton, D.A., Bowles, R.L. (2000). A windshear hazard index. *9th Conf. on Aviation, Range and Aerospace Meteorology*, Orlando, Amer. Meteor. Soc., 482–487.

[6] Shun, C.M., Chan, P.W. (2008). Applications of an infrared Doppler lidar in detection of wind shear. *J. Atmos. Oceanic Technol.* 25(5), 637–655. doi:10.1175/2007JTECHA1057.1

[7] Chan, P.W., Hon, K.K., Shin, D.K. (2011). Combined use of headwind ramps and gradients based on LIDAR data in the alerting of low-level windshear/turbulence. *Meteorol. Z.* 20(6), 661–670. doi:10.1127/0941-2948/2011/0242

[8] Hon, K.K., Chan, P.W. (2021). Improving lidar windshear detection efficiency by removal of "gentle ramps". *Atmosphere* 12(11), 1539. doi:10.3390/atmos12111539

[9] Cole, R.E., Todd, R.F. (1996). A comparative performance study of TDWR/LLWAS 3 integration algorithms for wind shear detection. *Workshop on Wind Shear and Wind Shear Alert Systems*, Oklahoma City, Amer. Meteor. Soc. (MIT Lincoln Laboratory).

[10] Haverdings, H., Chan, P.W. (2010). *Quick Access Recorder (QAR) data analysis software for studies of windshear, turbulence and wake vortex*. NLR-TP-2010-435, National Aerospace Laboratory NLR.

[11] ATC Network (2012). Finavia enhances surveillance capabilities through COMSOFT WAM network. 13 March 2012.

[12] Frequentis (2022). Finland transforms air traffic surveillance with Frequentis' highly resilient Wide Area Multilateration system. Press release, 20 June 2022.

[13] de Haan, S. (2011). High-resolution wind and temperature observations from aircraft tracked by Mode-S air traffic control radar. *J. Geophys. Res.* 116, D10111. doi:10.1029/2010JD015264

[14] Baas, P., Bosveld, F.C., Klein Baltink, H., Holtslag, A.A.M. (2009). A climatology of nocturnal low-level jets at Cabauw. *J. Appl. Meteor. Climatol.* 48(8), 1627–1642. doi:10.1175/2009JAMC1965.1

[15] Kopeć, J.M., Kwiatkowski, K., de Haan, S., Malinowski, S.P. (2016). Retrieving atmospheric turbulence information from regular commercial aircraft using Mode-S and ADS-B. *Atmos. Meas. Tech.* 9, 2253–2265. doi:10.5194/amt-9-2253-2016

[16] Ilmatieteen laitos (Finnish Meteorological Institute) (2021). *Lentosääpalvelut Suomessa* (Aviation weather services in Finland), 01/2021: §2.1.5 (WS in METAR), §2.1.6 (no SPECI), §2.3.3 (ARS and WXREP).

[17] MODE-S Wind project (2026). *Low-level windshear from passively received Mode-S EHS: literature status, a review of the six algorithms and a noise-floor and false-alarm test on 3 084 Helsinki-Vantaa landings*. Internal study, 24 October 2026 (with its literature collection, including Hong Kong, China: Deriving and using wind and temperature data from Mode S DAP and ADS-B data, ICAO APAC MET SG/27 IP/10, 2023).